LUMINE STARWORKS

THE CONNECTED ERA

2006 - 2026

MASTER WORLD BIBLE + FULL-STACK PRODUCT SPECIFICATION

Anime | Animation Industry | Web | Desktop | Browsers | Manga | Video | Music | Games | Offline Life

EDITION 1.0 | RESEARCH & IMPLEMENTATION BASELINE | 25 OCTOBER 2026

A complete specification of the product surface and technical architecture. Unfinalized franchise details are retained as explicit editorial decisions, never invented facts.

CHAPTER 01 / MASTER REFERENCE

Executive charter

What this project is, what “complete” means, and what has to be true for any component to ship.

Mission and central design promise

The Connected Era is Lumine StarWorks’ interconnected historical entertainment and internet environment covering 2006–2026 inclusive. It joins original anime and manga, animation studios, music, video sharing, social platforms, operating-system experiences, computer software, video games, and ordinary offline spaces. The user must be able to inhabit a selected date rather than merely read about it.

The initial build focus is summer 2012 through autumn 2013. Broader years remain part of the master design, but are added only after the first period functions. The 2013 YAL pages and the 2012 OFFLINE // OFFLINE computer are the principal initial reference anchors.

Definition of 100% for this book

- 100% of planned product classes have explicit functional contracts, input/output behavior, data ownership, persistence, permissions and acceptance tests.
- 100% of major architecture decisions have a selected default implementation or a transparent decision gate with options and consequences.
- 100% of established franchises and named products found in the consulted source set have a presence in a source-labelled registry, without fabricating unknown dates or staff.
- 100% of initial 2012–2013 end-to-end workflows are specified, including failure and offline paths.
- An open canon ledger remains legal and required. “100% spec coverage” does not mean every manga page, user post, scene or 2026 production credit has already been written.

Products inside one programme

Product family	User experience	First practical slice
Connected Desktop	Boot/log in, manage windows, applications, files and sessions.	Persistent 2012-2013 PC desktop.
Connected Browser	Navigate internal websites and permitted live sites.	Classic Firefox-/Chrome-era window skins.
Connected Search	Indexed search of fictional websites, videos, images and documents.	Full-text 2013 searchable catalogue.
YAL	Anime/manga list, catalog, comments, reviews, staff and studios.	Impact of Eden episode discussions and earlier catalog.
MeTube	Playback, profiles, comments, playlists, channels and uploads.	2012 home video and 2013 fandom clips.
Manga Reader	Read original/licensed manga, navigate, bookmark, synchronize.	Volume 1 chapter viewer.
OFFLINE // OFFLINE	Playable physical 2012 world with a computer and persistent NPC context.	Greg: July 11-12, 2012.
Culture Archive	Temporal records of offline venues, retail, memes, radio and events.	October 2013 localized snapshot.

Programmatic design constraints

- All original fictional services are functional applications, not mere static screenshots.
- Historical observation is separate from live modern participation.
- Use modern security underneath historical interface skinning.
- No commercial anime or manga reproduction without rights; original assets or authorized uploads only.
- No secrets or API keys in documents, source repositories, screenshots or seed databases.
- Respect the story/source continuity locks; alternative drafts remain labeled until approved.

Editorial status convention

[EST] = established in user source or explicitly finalized conversation; [WORK] = working design proposal; [OPEN] = unresolved canon or business decision; [EXT] = verified external technical/historical fact. These tags are carried through the book.

CHAPTER 02 / MASTER REFERENCE

Evidence, sources, and historical truth

Source provenance is required for every factual record and for every imagined historical reconstruction.

Four layers of truth

Layer	Example	Rules
Verified external history [EXT]	Firefox 24 shipped Sep 17 2013.	Attach official citation, geographical scope, and effective date.
Established fictional history [EST]	Impact of Eden regular TV premiere Oct 5 2013.	Attach specific authoritative franchise record or user lock.
Reconstructed cultural sample [WORK]	A fictional 2013 MeTube upload, reply or store sale.	Label as constructed, supply plausible date and format, no fake provenance.
Interactive player state [STATE]	A modern player adds a bookmark while exploring 2013.	Save independently of archival truth and timestamp with session clock.

Provenance fields

- Every franchise datum: subject ID, predicate, value, source document, source revision, author/decision owner, observed timestamp, lock status.
- Every external historical assertion: source URL, publication date, retrieval date, location when it matters, quotation/paraphrase policy.
- Every simulation record: publication date, created-at, effective start/end, audience visibility, actor ownership, reconstruction flag, moderation status.
- Every conflict: old value, new value, underlying reason, precedence rule and explicit resolution date.

Temporal integrity: five separate clocks

Field	Meaning	Critical behavior
occurred_at	When the event happened in the fictional timeline.	Premiere time is different from article time.
published_at	When a record became public.	Cannot be visible before it is published.
captured_at	When a historical screen or count was archived.	Oct 9 aggregate count not shown on Oct 6.
imported_at	When source was added to Lumine database.	Never determines historical visibility.
session_at	Time chosen by the explorer.	Advances available content but does not rewrite history.

A 2013 page replay can be both authored today and historically placed in 2013. The interface must state this is a reconstruction in its provenance/credits system; the on-screen story world can still use period-appropriate wording.

Early-scope authority ordering

- 1. Explicit latest user-approved locks override older alternatives.
- 2. Revised named “fixed” broadcast masters supersede older broad lore summaries where the scope matches.

- 3. Dated character/production corrections override legacy project naming.
- 4. Primary source documents outrank assistant-generated brainstorming when not explicitly finalized.
- 5. Missing content is displayed as pending rather than auto-completed.

CHAPTER 03 / MASTER REFERENCE

The 2006-2026 historical calendar

Year-by-year art direction, software behavior, culture and production gates. These are product-era design profiles, not a claim that every proposed platform existed in every year.

2006 | Personal-site transition

Cultural frame: Profiles, customized pages, fan signatures, web forums and downloadable MP3s are prominent reference cues.

Expected interactive affordances: Simple search pages, message boards, saved local files, dial-up/broadband variability.

Established/working fictional production anchors: Flow-era television and young LUNE debut form contextual media records.

- Date gate: at 2006-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

Editorial note: year framing above is a product-design interpretation; confirm any narrow historical statement against the external source register. For 2012-13, primary checkpoints are Microsoft, Mozilla, Google Chrome Releases and Pew research [W01-W06].

2007 | User video and creator identity

Cultural frame: Growing homemade uploads and fandom editing; individuals customize icons, forum signatures and playlists.

Expected interactive affordances: Video page with buffering, low-res streams, channel visits, manual recommendations.

Established/working fictional production anchors: Skyfall Chronicle late run, Iron Halo Brigade, Neon Afterimage and Tearfall Resonance.

- Date gate: at 2007-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2008 | Social browsing expansion

Cultural frame: Blogs, video comments, niche communities, media forums and stronger creator discovery.

Expected interactive affordances: RSS reader, threaded comments, simpler photo albums and favorites.

Established/working fictional production anchors: Circuit Hearts: Reload and Steelbound Testament among back-catalog.

- Date gate: at 2008-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2009 | Desktop social routines

Cultural frame: Persistent browser tabs, downloadable media, blogs, forums and increasingly everyday networking.

Expected interactive affordances: Vista/Windows 7 transition options, contacts, video playlists, notification trays.

Established/working fictional production anchors: Pale Garden Reverie, Nocturne of the Iron Sea, Steelbound Testament.

- Date gate: at 2009-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2010 | Fan communities converge

Cultural frame: Forum culture coexists with social networks, videos, fan fiction and hobby websites.

Expected interactive affordances: Image hosting, blog themes, group communities, feed experiments.

Established/working fictional production anchors: Blizzard Requiem and Garden of Cinders in historical catalog.

- Date gate: at 2010-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2011 | Smartphone growth and fandom overlap

Cultural frame: Portable communication accelerates while desktop browsers remain central in teen rooms.

Expected interactive affordances: Mixed mobile and desktop views; photos, chat and profiles.

Established/working fictional production anchors: Sinful Apex enters its original broadcast run.

- Date gate: at 2011-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2012 | Lived-in summer internet

Cultural frame: Greg’s July routines: personal computer, YAL, MeTube, contacts, downloaded music, local stores.

Expected interactive affordances: Windows 7-era desktop; downloads, buddy lists, old-style forums, physical manga.

Established/working fictional production anchors: Sinful Apex concludes; Rust Veil Junction and Skyfall Chronicle film records.

- Date gate: at 2012-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

Editorial note: year framing above is a product-design interpretation; confirm any narrow historical statement against the external source register. For 2012-13, primary checkpoints are Microsoft, Mozilla, Google Chrome Releases and Pew research [W01-W06].

2013 | Anime season and everyday web

Cultural frame: Community premiere reactions, forum quoting, AMVs, opening themes, local retail and video edits.

Expected interactive affordances: Firefox 24, Chrome 30 look; live profile updates, actual episode forums.

Established/working fictional production anchors: ITR season 1 ends Sep 26; IoE regular premiere Oct 5.

- Date gate: at 2013-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

Editorial note: year framing above is a product-design interpretation; confirm any narrow historical statement against the external source register. For 2012-13, primary checkpoints are Microsoft, Mozilla, Google Chrome Releases and Pew research [W01-W06].

2014 | Responsive web and mobile crossover

Cultural frame: Desktop forums coexist with rapidly growing mobile sharing and short-form social communication.

Expected interactive affordances: Websites may receive incremental responsive redesigns; preserve old profiles.

Established/working fictional production anchors: IoE continuity and Letters Under the Observatory Roof.

- Date gate: at 2014-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2015 | Bigger creators and streaming

Cultural frame: Algorithmic discovery grows, but standalone sites and loyal community groups persist.

Expected interactive affordances: Activity feeds, queues, improved media uploading, mobile companion views.

Established/working fictional production anchors: IoE 2014-15 continuity; early backlog and soundtrack growth.

- Date gate: at 2015-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2016 | Streaming and community acceleration

Cultural frame: Broadband-first video, creator channels and mobile fan communities.

Expected interactive affordances: High-res streams, adaptive playback, community video premieres.

Established/working fictional production anchors: EMC S1, The City That Forgot Its Lines.

- Date gate: at 2016-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

Editorial note: year framing above is a product-design interpretation; confirm any narrow historical statement against the external source register. For 2012-13, primary checkpoints are Microsoft, Mozilla, Google Chrome Releases and Pew research [W01-W06].

2017 | Creator networks

Cultural frame: Multi-platform creators, reactions, fan archives, mobile-first habits.

Expected interactive affordances: Subscriptions, threaded mentions, watch history, creator analytics.

Established/working fictional production anchors: ITR S2 Part 1, Bells Beneath the Last Station, Desk Kingdom No. 4.

- Date gate: at 2017-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2018 | Reaction and clip circulation

Cultural frame: Content crosses forums, video platforms, social feeds and offline events rapidly.

Expected interactive affordances: Clips, quote posts, GIF galleries, user collections.

Established/working fictional production anchors: ITR S2 Part 2, Salvagers, Spare Classroom and Radio Club.

- Date gate: at 2018-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2019 | Mature streaming era

Cultural frame: Multiple paid and ad-supported platforms, cine-anime promotion and dedicated fandom video.

Expected interactive affordances: Profiles, streaming queues, fan wikis, live-chats for scheduled events.

Established/working fictional production anchors: EMC S2, Overture, Velocity Crown and Orchard short film.

- Date gate: at 2019-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2020 | Remote-connected culture

Cultural frame: Online gatherings, watch parties, live video culture and complex communication flows.

Expected interactive affordances: Group watch rooms, direct messaging, remote community events.

Established/working fictional production anchors: ITR S3 premiere, EMC S3 and Moonphase Riot S1.

- Date gate: at 2020-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

Editorial note: year framing above is a product-design interpretation; confirm any narrow historical statement against the external source register. For 2012-13, primary checkpoints are Microsoft, Mozilla, Google Chrome Releases and Pew research [W01-W06].

2021 | Hybrid digital communities

Cultural frame: Fans circulate between live chat, uploads, fandom archives and offline return.

Expected interactive affordances: Community calendars, synced playlists, event announcements.

Established/working fictional production anchors: Battery Nocturne, Pocket Money Pilgrimage, IoE S4.

- Date gate: at 2021-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2022 | Creators and platform fragmentation

Cultural frame: Short clips, long videos, private servers, fan forums and direct chat coexist.

Expected interactive affordances: Cross-post metadata and new recommendation models.

Established/working fictional production anchors: Moonphase Riot S2, Rent Is Due After Sunset, IoE film/continuations.

- Date gate: at 2022-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2023 | High-volume digital media

Cultural frame: Mobile creators, production fandoms, streaming libraries and recommendation feeds.

Expected interactive affordances: Personal feeds, advanced search and historical archive views.

Established/working fictional production anchors: ITR S4, EMC S4, Tomorrow’s Headline, Moonphase Riot S3.

- Date gate: at 2023-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2024 | Legacy and live fandom side by side

Cultural frame: Older series regain attention through clips, retrospectives and social community memory.

Expected interactive affordances: Anniversary pages, archival comparison, metadata revision history.

Established/working fictional production anchors: Maps for Places That Move, Moonphase Riot production era.

- Date gate: at 2024-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2025 | Archive-conscious internet

Cultural frame: User interest in old internet aesthetics accompanies contemporary platform behaviors.

Expected interactive affordances: Switchable date skins, opt-in feeds, provenance and media history.

Established/working fictional production anchors: Moonphase Riot S4; EMC Journey’s End planned record.

- Date gate: at 2025-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

2026 | Algorithmic and AI-mediated landscape

Cultural frame: Generative media and platform consolidation coexist with deliberate archival reconstruction.

Expected interactive affordances: Transparent AI-content tags, preservation tools, searchable cross-era histories.

Established/working fictional production anchors: Moonphase Riot S5 schedule; LUNE twentieth anniversary; Garden at End of Orbit.

- Date gate: at 2026-01-01 00:00 local, only prior publications and scheduled releases are visible; within the year, time-of-day and publication timestamp still determine availability.
- Visual authenticity checkpoint: verify typography, screen density, browser chrome, media quality, slang, device penetration, and default discovery pattern against real period references.
- QA checkpoint: open a year-specific browser session, complete one search, read one media page, download one authorized asset, send one message, and verify persistence.

Editorial note: year framing above is a product-design interpretation; confirm any narrow historical statement against the external source register. For 2012-13, primary checkpoints are Microsoft, Mozilla, Google Chrome Releases and Pew research [W01-W06].

CHAPTER 04 / MASTER REFERENCE

Functional desktop operating environment

A simulated historical PC that has real state and real user actions, rather than a static imitation of a desktop.

Functional window manager

- Boot screen, sign-in and per-profile desktop restoration.
- Window open/minimize/maximize/resize/close; keyboard focus and Alt+Tab equivalent.
- Taskbar start menu, pinned programs, recent apps, clock, tray, network indicator, volume, notifications.
- Desktop icons, drag/reorder, grid snapping, wallpaper choice, screen resolution simulation.
- Multi-window session restore and crash recovery with durable window state.
- Settings: desktop theme, sound, date format, accessibility, privacy, installed apps, app defaults.

File system

- Virtual folders: Desktop, Documents, Downloads, Pictures, Music, Videos, Games, Recycle Bin.
- CRUD: create, rename, move, copy, duplicate, delete, recycle, restore, sort, search and inspect properties.
- Content-addressed blob objects; logical path table; unique IDs even after rename.
- Asynchronous download lifecycle with queued, active, paused, failed, finished, canceled.
- File associations launch media viewer, MP3 player, text editor, PDF viewer and archive manager.
- Exports to host filesystem occur only after explicit user choice; never silently access real local files.

User and session boundaries

Provide three distinct identities: (a) modern Lumine account, (b) historical reconstructed personas, (c) in-game NPC/character accounts. A modern user must not impersonate an established 2013 archival poster. Replay actions may be sandboxed or represented as modern annotations.

Cross-app state examples

Action	Write path	Consumers
Download wallpaper	file_object + download_job + folder_entry	Explorer, Photo Viewer, Desktop Settings
Bookmark an anime page	browser_bookmark + URL + session profile	Browser, YAL, desktop quick-links
Receive chat attachment	message_attachment + file_object	Messenger, Downloads, Photo Viewer
Listen to MP3	playback_state + playlist_item	Music Player, volume tray, Recent Files
Restart PC	persisted desktop_session snapshot	Window manager, notifications, app launchers

Desktop test scenarios

- D-01: Create “Anime Wallpapers” folder; download a file; move it; set wallpaper; restart and verify state.
- D-02: Open three apps, minimize one, send a message, restore and verify notification sequence.
- D-03: Attempt to open unknown file type; show a safe association dialog and avoid execution.
- D-04: Drag a 100MB file to Trash then restore it; content hash remains unchanged.

- D-05: Attempt to import a host folder without permission; access must be denied.

Historical operating system brand

A Windows 7-inspired visual experience is appropriate for the 2012 scene; shipping Microsoft’s original Windows OS or copyrighted system resources would require separate licensing. This specification designs a functional simulation shell, not a cloned Windows distribution. [W02, W03]

CHAPTER 05 / MASTER REFERENCE

Browsers, navigation and live web

Functional tabs and browsing history plus real web access only where permissions and sandbox boundaries allow.

Browser presentation variants

Skin / state	2012-2013 design	Runtime
Classic Fox-style	Top browser controls, older toolbar, add-ons, bookmarks/sidebar, download list.	Modern Chromium/Electron renderer.
Classic Chromium-style	Tabs, omnibox, new tab grid, starred bookmarks, menu, history.	Modern Chromium/Electron renderer.
Simple browser / school PC	Restricted navigation and sparse toolbar for scene authenticity.	Same engine, different policy and theme.

For an Oct 5 2013 reference, Firefox 24 was released Sep 17 and Chrome 30 entered stable Oct 1. Firefox 29 changes belong in 2014 [W04-W06].

Browser must genuinely support

- Address bar with typed scheme and full URL parsing; separate search fallback; Back/Forward/Reload/Home.
- Multiple tabs; duplicate, pin, mute, close and restore closed tabs; keyboard shortcuts.
- Distinct bookmark folders, favorite stars, browsing history, recently closed and per-session restoration.
- Find in page, download manager, progress/error states, HTTPS indicator, page zoom, font size and print-to-PDF.
- Integrated original services resolve to stable internal domains or registered secure app protocol.
- External sites open via sandboxed isolated web-content views; if blocked by CSP or disconnected, show clear explanation or open in system browser.
- No extension sideload or arbitrary script execution in first release; legacy Flash behavior only inside authorized sandboxed emulation.

Navigation security contract

- Maintain allowlisted navigation and permission prompts for camera, microphone, clipboard, notifications, geolocation and downloads.
- Never enable Node integration for remote pages; contextIsolation on and sandbox on; Electron recommendations [W07].
- Treat website title, icon and downloaded filename as hostile input; sanitize UI and filenames.
- Disallow file:// navigation to the host computer, automatic navigation to untrusted custom protocols and unrestricted popups.
- Explicit user action for links from internally fictional sites to the current live internet.

Search vs address behavior

Input “yal.connectedera.local/anime/impact-of-eden” routes internally; “impact of eden ep1” searches the period index. A fully-qualified external HTTPS URL opens via the allowed live-web path. Browser history stores origin, selected historical session date and actual access time separately.

Acceptance

- B-01: Four tabs can navigate independently, retain titles and survive restart.
- B-02: Bookmark a YAL entry, follow to MeTube, then return; history and forward navigation are correct.
- B-03: Outside website forbids embedding; viewer provides a safe external open option, never blank-fails silently.
- B-04: Custom protocol malicious payload cannot trigger host commands.
- B-05: In 2013 skin, 2014+ chrome appearance is disabled.

CHAPTER 06 / MASTER REFERENCE

Search engines and discovery

An operational search service for the fictional web, including pages, videos, manga, images, users, forums and archived records.

Search experiences

- General Search: typed query, phrase search, typo tolerance, snippet and dated ranked results.
- Images: thumbnail grid, origin credit, authorized downloads, reverse lookup limited to permitted assets.
- Video: duration, creator, upload date, views, content tags, playback link.
- Anime & Manga: franchise, genre, alternate titles, studios, episodes and staff.
- Forum: subject, reply contents, author, category, post date, thread pagination.
- Historical Results: filter publication date <= selected clock; never leak late retroactive commentary as contemporary truth.

Indexing and ranking

Start with PostgreSQL full-text indexing, trigram matching and a denormalized search_document table. Index source object ID, type, normalized title/body, publication date, canonical URL, visibility, language, keywords, media references and reindex version. Move to a dedicated search service only when measured latency or scale justifies it.

Stage	Scoring requirement	Guardrail
Keyword retrieval	Exact title and normalized phrase > fuzzy matches.	Exclude unlicensed or unavailable content.
Historical filter	published_at <= session_at and effective range includes date.	No future spoilers.
Relevance ranking	Text relevance + bounded period-appropriate popularity.	Avoid automatic 2026 modern boosts in 2013.
Result presentation	Title, link, snippet, date and source type.	Show broken/deleted statuses explicitly.
Click tracking	Anonymized engagement event with separate modern timestamp.	Do not alter historical popularity.

Special historic web conventions

- Queries should accept low-quality spelling, abbreviations, old forum syntax and language variants.
- Suggested results come from content present by that historical date.
- Image search supports 2013-style dense thumbnails, size filters and credible content ownership labeling.
- Archive 404 pages, domain moves and website shutdowns become modeled events.
- External current Google Search is a clearly labeled modern service, never passed off as a 2013 index.

Test scenarios

- SE-01: 2012 search finds manga volumes and older anime but not Oct 2013 episode threads.
- SE-02: Oct 6 search finds YAL ep1 thread after 02:31 JST, no Oct 13 ep2 thread.
- SE-03: Exact phrase and typo variant both recover the same original video.
- SE-04: Reported/deleted posts vanish from public search after moderation takes effect.

CHAPTER 07 / MASTER REFERENCE

MeTube: functioning video-sharing system

A distinct 2012-2013 homemade-video environment with complete upload, playback, channel and community state.

Media ingestion and ownership

- Upload video files (MP4/H.264 initially; other formats via server transcoding where available).
- Validate MIME by inspecting file contents; restrict limits; malware scan and moderation checks before publishing.
- Generate multiple encodes/resolutions, thumbnails and poster, durations and captions.
- Persist creator ID, title, description, tags, upload timestamp, visibility, licensing and source.
- Publish only videos created by Lumine StarWorks or users who hold the necessary upload rights.
- Provide private/unlisted/public modes; scheduled release date; revision and takedown history.

Video player

- Play, pause, seek, mute, volume, playback speed where applicable, keyboard controls, fullscreen.
- Provide optional dated buffering/loading design; never sabotage functional playback for nostalgia.
- Quality selection and adaptive bitrate when multiple renditions exist.
- Optional subtitles with WebVTT and accessible track labels.
- Watch-next and suggested list, channel info, comment count and uploader link.
- Proper handling for unavailable video, canceled upload, muted/removed rights and restricted age ranges.

Channels and social behavior

- Creator profile, banner, avatar, uploads, subscribers, playlists, featured video, description and creation date.
- Comment, edit own comment, reply, like, report and sort; chronological period accuracy.
- Favorites, subscriptions, watch later, playlists, video history and search.
- Creator dashboard: uploaded videos, engagement history, comments and rights status.
- Moderators can remove, restore with justification and lock comments.

Historical original creator seed

The OFFLINE // OFFLINE world bible names XXICEBURNERXX, HEARTCORE_ANGEL, NOOBDESTROYER92 and STRAWBERRY_STATIC. The seed may include period-style videos about 2012 soccer, mall failures, nightcore mixes and talent-show mishaps, only where explicitly marked as constructed reconstruction [F01].

Content performance targets

- A playable 10-minute video loads under normal development conditions and seeks without losing audio sync.
- The same video ID resolves from Search, MeTube channel page and Messenger shared link.
- View counters distinguish raw events, unique view approximation and captured historical snapshots.

CHAPTER 08 / MASTER REFERENCE

Manga, comics and digital reading

Reading manga must be a real sequence of page assets, bookmarks, chapters and reading state, not a gallery mockup.

Reader features

- Catalog browse by manga, creator, series, volume, chapter, genre, publication date and reading progress.
- Left-to-right and right-to-left mode; single/double page view; fit width/height; zoom; pan and keyboard navigation.
- Bookmarks at page, chapter and volume levels; last-read progress across sessions.
- Preloading of upcoming authorized page assets, offline caching when content license allows.
- Accessible chapter lists, alternative text for supplementary panels and OCR text layer only if supplied/authorized.
- User collections, ratings and recommendations independent from page reading.
- Page access control for ownership/subscription/preview tiers.

Data model

Entity	Minimum fields
series	series_id, title, alternate titles, author, genres, origin, status
volume	volume_id, series_id, number, publish_at, cover_asset_id
chapter	chapter_id, volume_id, order, title, publish_at, pages
page	page_id, image_asset_id, sequence, width, height, rtl_preference
reading_state	user_id, chapter_id, page, started_at, updated_at, completed_at
license	work_id, owner, permitted regions, use, expiry, proof reference

Specific OFFLINE story integration

On Thursday July 12, 2012, Lexi lends Greg Impact of Eden Volume 1. The manga is a persistent physical object, and Greg may leave it unread, read a little, read more, or read for most of the evening. Knowledge affects later conversations; no forced romance or destiny trigger [F02]. The reader may reproduce that optional experience digitally while keeping the physical-book continuity.

Publication and rights constraints

Do not scrape or republish commercial manga images from third-party websites. Seed original manga pages only when created/authorized. User-supplied imports stay private by default. The user can import authorized CBZ, PDF or image sequences into a personal local library with validation; sharing requires appropriate rights.

Acceptance

- M-01: Complete 24-page sample chapter and reopen at page 17 after application restart.
- M-02: Toggle right-to-left and observe navigation ordering change without corrupting progress.
- M-03: Chapter published later than current session date is unavailable in historical view.

CHAPTER 09 / MASTER REFERENCE

YAL: anime, manga and community archive

The first full implementation priority, based on the established teal-and-cream Impact of Eden forum references.

Historical identity

YourAnimeList is a traditional compact early-2010s anime/manga database. Primary top links: Anime, Manga, Community, Forums and Industry. Search field: “Search anime, manga, users...”. Main palette is dark brown-black #28231f, teal #2d7c77, warm page #eee8dc, cream #fffdf7, gold #e0aa35 and orange-brown #c0612c [F03].

Required pages and user actions

- Homepage with seasonal listings, editor posts, recent reviews, forums and search.
- Anime detail with title, aliases, studio, staff, genres, broadcast, episode list, score distribution, rank, users, related works, staff and characters.
- Manga detail with volumes, chapters, creator, reader links and reading-list actions.
- User profile with avatar, favorites, join date, anime list and forum history.
- Personal lists with Watching/Completed/On-Hold/Dropped/Plan to Watch, score 1-10, progress and notes.
- Forums: boards, topic creation, post, quote, spoiler, BBCode, report, follow, paginate and sort.
- Industry pages for production studio, directors, music, licensors, broadcasters and independent staff careers.
- Recommendations, member reviews with spoilers, favorites, rankings and statistical date snapshots.

Impact of Eden historical reference points

Event	Date/time	Rule
Episode 0 special	2013-09-26	Own special record and provenance.
Advance premiere	2013-10-02	Distinct from regular television premiere.
Regular premiere	2013-10-05	Series visible as currently airing.
YAL Episode 1 forum	2013-10-06 02:31 JST	Thread exists only from creation time.
YAL Episode 2 forum	2013-10-13	Thread appears on actual creation clock.
YAL Episode 3 forum	2013-10-20	Do not show before then.
Ep 1 replies = 1,113	Oct 9 archive snapshot	Never pass off later total as creation-day value.

IoE Season 1 is 25 episodes in the selected broadcast master. Staff: Kaihatsu Miura, Keito Kisaragi, Lio Haraguchi, Tomoe Suzuhara, Naochi Sumikuro, Mirai Kanzaki, Aoi Tachikaze, Yusuke Kagura and Hiroyuki Sawano. Main current names include Sora Gauthier, Abigail June, Grayson Rothenberg and Miyuki Kagemitsu. Legacy superseded names are not presented as current primary titles.

Episode forum visual contract

- Dark top bar with YAL logo and compact category links.
- Breadcrumbs, forum category trail, title card and metadata table.

- Old/new sort, first/previous/next page links, follow, reply and episode navigation.
- Post layout with 168px-style avatar/user column and message column, alternating warm backgrounds.
- Join/post counts, user status, edited markers, quote cards, spoiler reveal and moderator rank.
- Reconstructed thread posts stored with source provenance and capture timestamp distinct from modern live replies.

Historical OVA

Impact of Eden: Nightpolis Winter (2005) is a separate OVA database item with initial date Dec 21, 2005; its page has Details, Characters & Staff, Episodes, Reviews, Recommendations, Stats and Forum. Unknown studio, episode count and voices remain empty until verified [F04].

Acceptance

- Y-01: Episode 1/2/3 screens match their supplied HTML reference structure.
- Y-02: 2012 profile displays manga content but not a future 2013 episode reaction.
- Y-03: Adding a list item persists between desktop and embedded YAL app.
- Y-04: Legacy aliases never overwrite final character names.
- Y-05: Search results and screenshot metadata are time-consistent.

CHAPTER 10 / MASTER REFERENCE

MunchyRoll, AniBlex, Funmaxation and broadcast media

The anime ecosystem needs independent distribution, streaming, television and promotional services.

MunchyRoll

Identity: Licensed and original anime streaming

Required working features: Series detail, video player, subtitle/language track, regional availability, watch queue, watch history, premium entitlements, episode schedule, opening-closing data, notifications.

Source and rights rule: Playback entitlements and schedules respect license, region and historical date.

AniBlex

Identity: Japanese production / licensing / promotion

Required working features: Production announcements, discography, trailers, press releases, merchandise, home-media SKU, limited events, release calendar, artist pages.

Source and rights rule: Separate producer, licensor, distributor and broadcaster roles.

Funmaxation

Identity: North American dubbing and distribution

Required working features: Dub casting, local premiere date, license region, physical distribution, streaming partners, audio format, press events.

Source and rights rule: Japanese broadcast and NA dub dates remain independent.

AdultDrift

Identity: Broadcast television schedule

Required working features: Channel guide, timezone-based listings, reruns, dub flags, content rating, promotional interstitials, archived programme guides.

Source and rights rule: A historical guide may replay information; no actual live rebroadcast without licensed video.

AniWire

Identity: Editorial anime journalism

Required working features: Dated articles, authors, media, tags, interviews, corrections, comments and related title links.

Source and rights rule: Corrections recorded as amendments; future review unseen prior to publication.

SakugaThread

Identity: Animation analysis community

Required working features: Cut breakdowns, credits, clip ownership, staff search, annotations, frame comparisons and comments.

Source and rights rule: Only use authorized sample clips and licensed stills.

Soniplex

Identity: Named production/distribution entity in older drafts

Required working features: Company record, project press releases, trailer announcement, staff/ownership links.

Source and rights rule: Relationship to AniBlex unresolved; do not merge until decided.

Broadcast Networks

Identity: Japanese TV and schedule archives

Required working features: Network profiles, station airings, cour scheduling, hiatus, reair, local time, programming block records.

Source and rights rule: Broadcast masters may conflict; source revision and network station must be captured.

CHAPTER 11 / MASTER REFERENCE

Social sites, chat, blogs and early-web communities

The web must feel heterogeneous: each service needs its own interaction and visual grammar.

Birdie

Functional contract: 140-character period-mode posts, profiles, followers, hashtags, replies, reposts, favorites, mentions, direct messages if supported; 2012-13 chronology is not a modern infinite-video feed.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Tum

Functional contract: Themeable blogs, post types, image sets, GIFs, reblogs, following, archives, ask-box, tagging, custom backgrounds and typography.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

3Chan

Functional contract: Anonymous boards, time-limited threads, OP images, reply IDs, quotations, board rules, deletion/archival, report tools; built-in safety controls.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Community Forum

Functional contract: Categories, subjects, threads, BBCode, classic avatars, signatures, pagination, moderator actions and older thread bump order.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Messenger

Functional contract: Buddy list, away/online/offline/invisible, group chat, timestamps, avatars, contact requests, file sharing and persistent history.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.

- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Email

Functional contract: Draft/sent/inbox, local fictional mailboxes, attachment controls, spam, search, reply/forward; optionally integrate real email separately via account authorization.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

IRC-like Chat

Functional contract: Join channel, nickname, topic, private messages, away state, moderation, scrollback and plain text export.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Personal Website / Guestbook

Functional contract: HTML-like templates, visitor count (simulated), profile music, banners, guestbook replies, curated link lists.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

RSS Reader

Functional contract: Subscribe to compatible feeds, unread counts, folders, fetch history, chronological updates, local caching.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Fan-fiction Archive

Functional contract: Work/chapter upload, tags, pairings/warnings where appropriate, comments, bookmarks, revision history and reader preferences.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Fan Art Gallery

Functional contract: Artwork upload, galleries, search tags, watermarks by choice, favorites, creator pages, safe viewer and moderation.

- Persistence: all updates survive refresh and returning to the desktop.

- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Photo Hosting

Functional contract: Albums, access controls, share links, slideshow, tags, title/description and downloadable authorized photos.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Local/School Forum

Functional contract: Community boards with fictional town events, clubs, sports and moderation; visible according to school calendar.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Marketplace / Classifieds

Functional contract: Original virtual listings, categories, seller profiles, messages and archived sold states; avoid real-money transactions in initial slice.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

Wiki and Fan Encyclopedia

Functional contract: Revisioned pages, history, talk tabs, infoboxes, search, editor permissions and spoiler handling.

- Persistence: all updates survive refresh and returning to the desktop.
- Time gate: archived history never accepts a backdated user post masquerading as an established period record.
- Safety: role-based moderation, reporting, deletion audit trail and content escaping are mandatory.

CHAPTER 12 / MASTER REFERENCE

Complete desktop software inventory

Functional legacy-feeling applications are integrated through common file, notification, settings and identity services.

01. File Explorer

Functional requirements: folders, tree/list/detail modes, copy/move/rename/recycle, breadcrumbs, search, previews. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

02. Download Manager

Functional requirements: save location, progress, retry/pause/cancel, scan verdict, open/reveal file. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

03. MP3 Music Player

Functional requirements: import local songs, playlists, seek, shuffle, repeat, album art and library scanning. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

04. Video Player

Functional requirements: local MP4 play, captions, speed, fullscreen, playlist, file association. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

05. Photo Viewer

Functional requirements: zoom, next/previous, rotate, simple slideshow, wallpaper selection. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

06. Paint / Image Editor

Functional requirements: brush, colors, fill, crop, layers optional, export PNG. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

07. Notepad

Functional requirements: open/edit/save text, encoding choice, find/replace, autosave optional. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

08. Calculator

Functional requirements: basic/scientific operations, copy result, keyboard support. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

09. Calendar & Reminders

Functional requirements: day/week/month, event CRUD, recurrence, reminder notifications. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

10. PDF Reader

Functional requirements: zoom, page navigate, text search, selection, bookmarks, print. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

11. Archive / ZIP Utility

Functional requirements: extract/create ZIP with traversal protections and size limits. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

12. File Properties

Functional requirements: type, size, owner, modification dates, source origin and rights metadata. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

13. System Settings

Functional requirements: time zone, display, audio, notifications, wallpaper, language, privacy. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

14. Recycle Bin

Functional requirements: restore, empty, retention policy, persistent deletion states. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

15. Task Manager

Functional requirements: simulated app processes, resource indicators, force close fictional apps safely. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

16. Text Chat Client

Functional requirements: contacts, groups, statuses, files, sounds and searchable history. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

17. Audio Recorder

Functional requirements: record permission-based microphone input, save WAV or supported codec. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

18. Webcam App

Functional requirements: camera permission, capture, local photos, safe default off. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

19. Screenshot Utility

Functional requirements: select window or desktop region, save authorized screenshots. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

20. Game Launcher

Functional requirements: installed games, save slots, achievements, local launch and compatibility. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

21. Wallpaper/Theme Gallery

Functional requirements: browse authorized assets, preview, apply, favorites, persisted themes. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

22. Media Converter

Functional requirements: convert permitted personal media to common audio/video formats. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

23. RSS / News Reader

Functional requirements: feeds, unread markers, folder subscriptions, offline cache. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

24. Maps / Local Directory

Functional requirements: fictional town locations, categories, opening-hours records, directions where map supports them. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

25. Clock / Alarm

Functional requirements: local date, historical session indicator, timers and alarms. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

26. Disc Collection Manager

Functional requirements: organize CD/DVD/media items, track loaned assets and metadata. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

27. Drawing Tablet Utility

Functional requirements: pressure input where available, canvas save, brush parameters. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

28. Simple Office Editor

Functional requirements: notes, lists, basic tables, print/export. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

29. Browser Game Host

Functional requirements: sandboxed Ruffle/HTML5 original games, persistent save scores. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

30. System Help

Functional requirements: searchable guides, user help, troubleshooting, recovery hints. The application must integrate with shared user identity, file selection, keyboard commands, error states and desktop window manager.

Save and verification: user modifications persist across reboot and re-open; app shows a meaningful empty state and refuses unsupported file types with a recoverable message.

CHAPTER 13 / MASTER REFERENCE

Browser games, gaming sites and social play

Playable games, launchers, achievements, community events and arcade experiences across eras.

Playable browser content

- Implement original HTML5 games as first-class hosted apps (race, puzzle, platformer, rhythm, quiz).
- Optional compatible user-authorized SWF titles via Ruffle. Unsupported Flash behavior is clearly labeled; no claim of 100% compatibility [W11].
- Each game has metadata: release, creator, version, genres, controls, save behavior, asset license, compatible era skin.
- Local scores persist in SaveData; leaderboard entries require an explicit online session and anti-cheat boundaries.
- Game page contains screenshot, trailer, controls, rating, comments, install/play button and discussion link.

Community gaming ecosystem

- Gaming news, reviews, walkthroughs, fan-created guides, patch notes, modding communities and clans.
- Console and PC game directories as historical *reference entries* without unauthorized redistribution.
- LAN party events, arcade tournaments, seasonal contests and school computer lab sessions.
- MeTube channels link to original gameplay clips; shared ID relationships avoid duplicate media.
- Player can bookmark guides and save permitted screenshots to desktop.

OFFLINE arcade specification

The July 12 2012 planning bible lists a racing cabinet, fighting game, light-gun shooter, air hockey, basketball machine, rhythm game, claw machine and ticket games. They are optional low-stakes activities; friend performance, spent money and prizes can cause later dialogue references, not heavy branching consequences [F02].

Success tests

- G-01: Launch original arcade racer; finish run; save high score; return next game session.
- G-02: Game guide opens in one browser tab while game remains paused in another.
- G-03: MeTube video from game directory is same asset reached by MeTube search.
- G-04: A 2018 title is absent in 2012 store inventory and search until its release clock.

CHAPTER 14 / MASTER REFERENCE

Offline cultural world: where internet meets the street

The Connected Era includes retail, conventions, transport, school routines and physical culture, not merely screens.

Bookstore & Manga Shop

Functional and content requirements: shelved volumes; stock availability; browsing; purchases; magazine release dates; promotions; borrowed book memories. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Shopping Mall

Functional and content requirements: directory, food court, stores, movie theater, background NPC routines, receipts, shopping bags. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Arcade

Functional and content requirements: working mini-games, scores, credits, tickets, rewards, ambient sound and local rivals. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

School/Club Rooms

Functional and content requirements: calendars, classes, school boards, clubs, tournaments, notices, yearbook content. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Conventions & Fan Meetups

Functional and content requirements: venues, ticket records, guests, panels, event schedules, merchandise, artist alleys, cosplayers. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Record/CD Store

Functional and content requirements: albums, posters, demo headphones, physical singles, local band promotions. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Electronics Retail

Functional and content requirements: desktop and smartphone models appropriate to the period, demo PCs, peripherals, game cases. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Local Cinema

Functional and content requirements: showtimes, ticketing mock (non-real payment), theatrical premieres and trailers. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Cafes and Food Court

Functional and content requirements: menus, opening hours, NPC chats, seating and purchases. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Neighborhood Streets

Functional and content requirements: ambient traffic, pedestrians, bikes, deliveries, benches, sprinklers and weather. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Local Radio/TV

Functional and content requirements: historical playlists, programme schedule, DJ voice lines, fictional advertising. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Public Library

Functional and content requirements: computers, paper books, lending records, local notices, silent reading spaces. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Parks and Recreation

Functional and content requirements: basketball court, soccer, skatepark, picnic, seasonal activities. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Part-Time Job Locations

Functional and content requirements: schedule, tasks, pay ledger, colleague relationships, everyday scenes. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Local Calendar

Functional and content requirements: school dates, holidays, concerts, sales, sporting events, convention weekends. Each location has time-aware availability, physical map position, interaction list, NPC schedule, and archive-appropriate media inventory.

Cross-channel cultural events

An anime single launch may affect AniBlex’s news page, a music CD shelf, a creator review on MeTube, radio station promos, a convention event and an NPC’s conversation. Each effect references one historical event ID and its own publication/display timestamp.

CHAPTER 15 / MASTER REFERENCE

OFFLINE // OFFLINE Worlds: continuity and implementation

The physical world in 2012, its cast, daily routines, user agency, and data linkage to Connected Desktop.

Established material [F01, F02]

- Episode 1: “Before the Day Starts” on July 11, 2012; Greg at home, court, store and park with Ren, Jay, Lexi.
- Greg lives with Mom; father/home-history detail is not established in the consulted bible.
- Greg is a soccer player, not a basketball player; friends tease one another with continuity.
- Ren’s slide photo, group chat “supahots”, and Lexi’s offer to lend Volume 1 persist into Episode 2.
- Episode 2: Thursday July 12, 2012; mall, bookstore, food court, arcade, afternoon split, dinner, late-night reading choice.
- Offline Mode and Online Mode are distinct game layers; two fixed IoE protagonists belong to Online Mode.

Episode 2 detailed interaction schedule

Time	Scene	Persisted consequence
07:30-09:00	Wake, messages, breakfast, home chore	Task, messaging, bedroom state
09:00-10:20	Neighborhood walk, meet friends, receive Volume 1	Inventory book, group context
10:20-12:30	Go to mall; browse bookstore	Town familiarity, item views
12:30-14:30	Food court and arcade	Choices, scores, money
14:30-16:30	Ren/Jay/Lexi/solo optional branch	Companion-specific memory
16:30-19:30	Home, computer, dinner	Browser, parent conversation
19:30-23:30	Free time, optional manga, message, sleep	Reading progress and day transition

World regions and source conflict

The Crescent Valley/Branford/unnamed-town conflict is explicit in the source bible. Crescent Valley is attached to an alternate Lucas/Mia/Jordan/Erin group and Lincoln High; Greg’s active core is Ren/Jay/Lexi. The spec therefore uses a neutral world_id and editable display name until a geography decision is made. Do not merge old friend groups into one universe by assumption [F01].

Persistent world state

- World time, day, weather, location unlocks and player position.
- Inventory (book, purchases, receipts), money and daily jobs.
- Message history, photos, device downloads, browser history, music, user profile status.
- NPC mood and dialogue memory, friend interactions, arcade performance, town familiarity.
- Branch choice consequences are light and human; no progression meters for friendship or shopping.
- Local stock, TV/radio content and billboards query the historical event database.

Acceptance

- O-01: Ren’s slide photo is available in July 12 chat and photo collection.
- O-02: Volume 1 persists when carried home; Greg can leave it unread.
- O-03: Two independent July 12 saves with different reading progress produce different later Lexi dialogue.
- O-04: Greg’s internet bookmarks from PC sessions persist between game days.
- O-05: Town label is not hard-coded before canonical decision.

CHAPTER 16 / MASTER REFERENCE

Anime franchise registry and production continuity

The known franchise anchors, broadcast installments and title-level data required by YAL and the broader industry archive.

Franchise	Release anchors	Established/working structure	Outstanding
Impact of Eden	2013 regular premiere Oct 5; Episode 0 Sep 26; advance Oct 2	Studio Burst; director Miura; score Sawano; initial TV S1 25 [master revision]	Full episode-total and later master conflicts remain.
Iron to Requiem	S1 Spring 2013; S2 Winter 2017 + Spring 2018; S3 Fall 2020; S4 Summer 2023	4 numbered seasons, 5 installments, 102 eps (26+12+15+24+25); S1 CHAOS Home	S2P1, S3 and S4 exact schedules working; later studio credits open.
EARTH: Metal Comet	2016, 2019, 2020, 2023 TV; Journey's End 2025 film	S1 12 + S2 22 + S3 12 + S4 25 = 71; Burst then FlareWorks	Dates in broader archive need final time-of-day and release-territory review.
Sinful Apex	Aug 15 2011 - Jan 23 2012	24 eps; Flow Pictures; LUNE as Aria Moonbirth	Detailed character/song map partly open.
Overture	2019 theatrical	Studio Burst; Sawano score; LUNE ending Goodbye, Past Identity	Precise premiere region / box office / runtime open.
Moonphase Riot	2020-2026 planned five seasons	24+12+24+12+12=84 eps, Studio Burst, Taku Iwasaki	2026 scheduled finale future; cross-check weekday/date issues.
Starfall upon Humanity	Era in Burst archive, exact title record incomplete	Burst chamber; Sawano-connected; LUNE associated	Full production ledger open.
ARATAKI	2027 future planned	Burst chamber; Yasuharu Takanashi score; LUNE proposed	Outside project release range; record placeholder only.

ITR installment detail and broadcast clocks

Installment	Date range	Ep.	Status
Season 1	Apr 4 - Sep 26 2013	26	[EST] 26 Thursdays
Season 2 Part 1	Jan 12 - Mar 30 2017	12	[WORK] precise dates
Season 2 Part 2	Apr 5 - Jul 12 2018	15	[EST] explicitly approved
Season 3	Oct 3 2020 - Mar 13 2021	24	[WORK] precise dates
Season 4	Jul 8 - Dec 23 2023	25	[WORK] precise dates

IoE current character naming

Use Sora Gauthier (not Nate), Grayson Rothenberg (not Sebastian where replaced), Kaito Umbracall where newer records apply, Abigail June and Miyuki Kagemitsu. Some old HTML and historic production drafts retain the former name; database supports a superseded_alias record but never labels it as current canon.

CHAPTER 17 / MASTER REFERENCE

Studio Burst filmography beyond the major pillars

All twenty archive titles deserve distinct database records, including audience, genre, season and eventually full production credits.

01. Impact of Eden

Release: 2013-2024. Format: TV + films. Catalogue premise: Adventurous supernatural science-fiction ensemble; flagship.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

02. Overture

Release: 2019. Format: Theatrical film. Catalogue premise: Sci-fi fantasy, music, action and emotional drama.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

03. EARTH: Metal Comet

Release: 2016-2025. Format: TV + film. Catalogue premise: Coming-of-age cosmic adventure; two-studio TV history.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

04. Letters Under the Observatory Roof

Release: Summer 2014. Format: 12 TV eps. Catalogue premise: Two students exchanging unsent letters; quiet school romance.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

05. The City That Forgot Its Lines

Release: Fall 2016. Format: 12 TV eps. Catalogue premise: Invisible city lines, psychological mystery and courier protagonist.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

06. Bells Beneath the Last Station

Release: Spring 2017. Format: 24 TV eps. Catalogue premise: Runaway teenagers, abandoned chapel, crime and radio-linked supernatural system.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

07. Desk Kingdom No. 4

Release: Summer 2017. Format: 12 TV eps. Catalogue premise: 4-koma desk territories, snacks and school chaos.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

08. Salvagers of the Falling Blue

Release: Winter 2018. Format: 24 TV eps. Catalogue premise: Scavengers explore collapsing floating continents.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

09. The Spare Classroom on Saturday

Release: Spring 2018. Format: 12 TV eps. Catalogue premise: A found family spending weekends in an unused classroom.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

10. The Radio Club Heard Everything

Release: Fall 2018. Format: 13 TV eps. Catalogue premise: School broadcast picks up emotional frequencies.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

11. The Orchard Where Machines Sleep

Release: 2019. Format: Short film. Catalogue premise: Old war machines reused in rural farm life.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

12. Velocity Crown

Release: Spring 2019. Format: 12 TV eps. Catalogue premise: Drone- and momentum-based futuristic team sport.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

13. Moonphase Riot

Release: Fall 2020+. Format: 5 TV seasons. Catalogue premise: Night patrol workers confront lunar reality anomalies.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

14. Night Shift Constellation

Release: Spring 2020. Format: 24 x 5-min shorts. Catalogue premise: Late-night convenience-store workplace comedy.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

15. Battery Nocturne

Release: Spring 2021. Format: 13 TV eps. Catalogue premise: Musicians stabilize threatened city districts with sound batteries.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

16. Pocket Money Pilgrimage

Release: Summer 2021. Format: 12 TV eps. Catalogue premise: Three middle-schoolers build an errand club.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

17. Rent Is Due After Sunset

Release: Spring 2022. Format: 13 TV eps. Catalogue premise: Young-adult apartment and workplace romance.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

18. Tomorrow’s Headline Is Missing

Release: Winter 2023. Format: 11 TV eps. Catalogue premise: Local newspaper prints future headlines with consequences.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

19. Maps for Places That Move

Release: Spring 2024. Format: 24 TV eps. Catalogue premise: Young mapmakers travel across shifting borderlands.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

20. The Garden at the End of Orbit

Release: Summer 2026. Format: 12 TV eps. Catalogue premise: Space colony garden recreates memories and lost ecosystems.

Required archive pages: main entry, release installments, staff and composer credits, related works, broadcast/publication timetable, media assets, ratings with captured dates, reviews, forum posts, soundtrack linkages and rights metadata. Untold details remain marked [OPEN].

CHAPTER 18 / MASTER REFERENCE

Flow Pictures earlier catalogue and creative inheritance

Production history before the 2013 Studio Burst launch gives the anime ecosystem decades of staff continuity.

01. Vacant Crown: Liminal War

Historical catalogue: 2000; TV + OVA. Core description: Supernatural weapons and wide-angle military conflict.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

02. River of Sparks

Historical catalogue: Winter 2004; TV. Core description: Delinquent school/urban magic, close-quarters fighting.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

03. Glass Cathedral

Historical catalogue: 2005; Film. Core description: Supernatural crime and stained-glass imagery.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

04. Skyfall Chronicle

Historical catalogue: 2006-2007; 2-cour TV. Core description: Floating citadels and warfare.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

05. Iron Halo Brigade

Historical catalogue: 2007; 1-cour TV. Core description: Military mecha and infantry, realistic tactics.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

06. Neon Afterimage

Historical catalogue: 2007; Film. Core description: Rain-soaked urban science fiction.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

07. Tearfall Resonance

Historical catalogue: 2007; Film. Core description: Siblings involved in a psychic city disaster.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

08. Circuit Hearts: Reload

Historical catalogue: 2008; 1-cour TV. Core description: Illegal neural-gear competitions.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

09. Steelbound Testament

Historical catalogue: 2008-2009; 2-cour TV. Core description: Dark fantasy, family and war.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

10. Pale Garden Reverie

Historical catalogue: 2009; 1-cour TV. Core description: Quiet supernatural environmental storytelling.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

11. Nocturne of the Iron Sea

Historical catalogue: 2009; Film. Core description: Naval horror-adventure.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

12. Blizzard Requiem

Historical catalogue: 2010-2011; Split 2-cour TV. Core description: Arctic survival and politics.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

13. Garden of Cinders

Historical catalogue: 2010; OVA trilogy. Core description: Steelbound Testament side story.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

14. Shadowline Oratorio

Historical catalogue: 2011; 1-cour TV. Core description: Music and assassin-choir school.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

15. Zero Latitude

Historical catalogue: 2011; TV special. Core description: Blackout megacity chase.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

16. Sinful Apex

Historical catalogue: 2011-2012; 24 TV eps. Core description: Final marquee Flow production.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

17. Rust Veil Junction

Historical catalogue: 2012; 1-cour TV. Core description: Steampunk rail city factions.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

18. Skyfall Chronicle: Black Aegis

Historical catalogue: 2012; Film. Core description: Theatrical sky-warfare sequel.

Credit caution: catalogue and pre-Burst staff histories must not be mistaken for a definitive animation-studio ownership list for every work. Store production-company confidence and source references separately.

Other established earlier staff-filmography titles

Ghost Eater, Destiny/Infinite, A Mystical Skyline/Finger and Azure Sorcerer appear as individual older production credits but their comprehensive broadcasting/studio records are not fixed in the consulted excerpts. Create their title IDs; avoid invented seasons or producers.

People registry

Person	Role	Archive connections
Kaihatsu Miura	Director / studio founder	Sinful Apex, Skyfall, Zero Latitude, IoE and EMC
Keito Kisaragi	Writer / composition	Steelbound Testament, Pale Garden, Blizzard Requiem
Tomoe Suzuhara	Storyboarding	Glass Cathedral, Vacant Crown, A Mystical Skyline
Lio Haraguchi	Character / animation	Destiny/Infinite, Steelbound Testament, Sinful Apex
Yusuke Kagura	Fight choreography	River of Sparks, Ghost Eater, Sinful Apex
Mirai Kanzaki	Color design	Sinful Apex, Neon Afterimage
Aoi Tachikaze	Sound direction	River of Sparks, Glass Cathedral
Rento Shibasaki	Music composition	Vacant Crown, Steelbound Testament

CHAPTER 19 / MASTER REFERENCE

Music, radio, theme songs and creator identity

All audio uses linked artists, works, recordings, releases, locations and listener activity records.

LUNE: documented trajectory

Lena Moonlight (LUNE) debuted professionally in 2006 at age 13, remained an independent singer-songwriter, had recurring Flow Pictures work, appeared as Aria Moonbirth in Sinful Apex, carried creative relationships to Burst, performed Goodbye, Past Identity for Overture and returned for Dear Tomorrow in 2026. Her exact credits for some works remain open [F07].

Project	Documented or working song
Vacant Crown: Liminal War	Gesshoku / Requiem for Zero
Glass Cathedral	VITRAIL
Neon Afterimage	AFTER//IMAGE
Circuit Hearts: Reload	PULSE:RE
Steelbound Testament	ALETHEIA / Ashes Know My Name
Overture	Goodbye, Past Identity
Impact of Eden	Dear Tomorrow (2026), World Gauntlet ED title open

Radio stations [F01]

- CV Pop FM: mainstream music and commercial DJs. Must have scheduled programming, episode-level broadcast dates and licensed/original track references.
- Garage Noise 101.7: indie, punk, alternative and emo. Must have scheduled programming, episode-level broadcast dates and licensed/original track references.
- Ambient Echo: relaxing ambient and electronic instrumentals. Must have scheduled programming, episode-level broadcast dates and licensed/original track references.
- TalkWave AM: local news, sports, gossip and talk. Must have scheduled programming, episode-level broadcast dates and licensed/original track references.
- Pixel Beats: gaming and internet culture, anime promotional cues. Must have scheduled programming, episode-level broadcast dates and licensed/original track references.
- Golden Oldies: older music and the parents’ listening habits. Must have scheduled programming, episode-level broadcast dates and licensed/original track references.

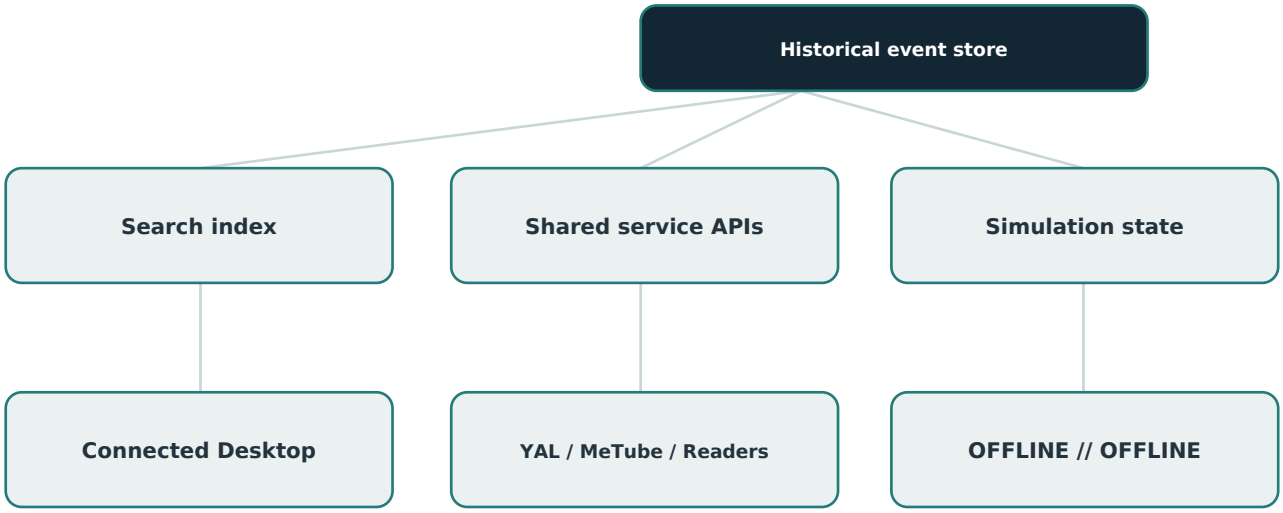
Audio functionality

- Master track registry: artist, performers, composition, recording, lyrics credit, label, release, file, rights.
- Play original/licensed audio with queue, station schedule, seek, local media player and radio streams.
- A radio broadcast has program_id + airing timestamp + cue list and must not leak future songs.
- MeTube may link to a song but must not duplicate music file ownership across services.
- Show credits: score composer is distinct from vocalist; do not call Sawano’s IoE work mere stylistic inspiration.

CHAPTER 20 / MASTER REFERENCE

Shared technical architecture

A secure implementation that preserves the period aesthetic while running on maintained software.



Architecture relationship, not network topology. All app surfaces query the same time-aware records.

Reference system shape

- Desktop shell: Electron + TypeScript/React for standalone PC, or web-only fallback using browser-window simulations.
- Internal sites: Next.js applications, shared design assets but independent content domains and public UI identities.
- Core API: TypeScript server modules and REST/RPC endpoints with explicit authorization.
- Database: PostgreSQL for canonical records and time-aware metadata; storage bucket for user-owned media and permitted content.
- Search: Postgres full-text for first release, pluggable provider behind search interface.
- Chat/presence: real-time channel system with persistence in relational storage.
- Local file cache: IndexedDB/OPFS for sandboxed browser mode; private app path in desktop mode.
- Events: append-only history for publication, moderation, release, migration and provenance decisions.

Product domains / namespaces

Use reserved development hostnames that cannot conflict with live commercial sites, e.g. yal.connectedera.local, metube.connectedera.local, manga.connectedera.local and search.connectedera.local. In Electron, translate them through a controlled custom protocol or trusted local development server. Do not assume that inventing a domain gives a public DNS registration.

Event bus

Publication events such as episode.premiered, post.published, video.uploaded, wallpaper.saved and store.stock_changed are immutable payloads with event_id, actor, occurrence time, publication time, source ID, effective time and transaction ID. Consumers may update search, notifications and UI projections idempotently.

Cross-service provenance and safety

- Never treat a client-provided historical timestamp as authoritative when accepting user-generated modern content.
- Historic scripted NPC messages are stored as simulated records with authored_at and in_story_at.
- Live user data is always separable and exportable; original reconstructed content is versioned.
- Staff corrections should cascade through linked entries via relations, not plain-text overwrites.

CHAPTER 21 / MASTER REFERENCE

Core data dictionary and relational schema

The following tables are a build-ready logical model. Physical schema, migrations and exact indexes can be generated from these field contracts.

IDENTITY TABLE FAMILY

- accounts: id, email/identity provider, created_at, suspended_at
- profiles: account_id, display_name, avatar, privacy, locale, timezone
- historical_personas: persona_id, period_start/end, provenance, control policy
- roles_permissions: subject, role, resource, permission, granted_at

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

CULTURE TABLE FAMILY

- history_events: id, type, title, occurred_at, published_at, captured_at, region
- source_documents: id, filename, revision, extraction location, owner, hash
- source_assertions: id, subject, predicate, value, source_id, confidence, valid_from
- canon_decisions: id, conflict, proposed resolutions, final value, approved_by
- temporal_snapshots: source_ref, observed_at, counts_json, publication_state

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

ANIME TABLE FAMILY

- franchises: id, title, creator, primary_language
- anime_entries: id, franchise_id, format, season, status, synopsis
- episodes: id, anime_id, sequence, title, aired_at, duration
- manga_series, volumes, chapters, pages
- characters: id, name, aliases, roles, debut_at
- persons, companies, staff_credits, character_voices
- release_windows: work, territory, network, start, end, language
- related_works: from_work, to_work, relationship, valid_time

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

WEB TABLE FAMILY

- websites: id, name, domain, debut_at, skin_versions
- site_pages: id, website_id, path, type, published_at, visibility
- search_documents: id, url, kind, tokens, published_at, modified_at
- browser_history: user_id, profile_id, URL, access_at, session_at
- browser_bookmarks: user_id, URL, folder, created_at
- site_accounts: user_id, service_id, profile_id

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

VIDEO TABLE FAMILY

- videos: id, owner_id, title, description, published_at, visibility, rights
- video_renditions: video_id, encoding, resolution, blob_id, duration
- channels, subscriptions, playlists, playlist_items
- video_comments, reactions, view_events, view_snapshots
- captions: video_id, locale, asset_id, format

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

FORUMS TABLE FAMILY

- boards: id, site_id, name, rules
- threads: id, board_id, title, opened_at, locked_at
- posts: id, thread_id, persona/user, published_at, content, quoted_post_id
- post_edits, thread_follows, reports, moderation_events

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

DESKTOP TABLE FAMILY

- device_profiles: user_id, device_id, era_skin, settings
- desktop_sessions: device_id, opened_at, closed_at, windows_json
- window_state: app, bounds, z_index, minimized, focused
- file_objects: id, hash, owner, content_type, storage_path
- file_entries: parent_folder, name, blob_id, deleted_at
- download_jobs: URL, user_id, bytes, status, attempt_count
- notification_events, local_media_playback, app_preferences

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

GAME TABLE FAMILY

- game_projects: id, title, formats, release
- game_sessions: player_id, game_id, started_at, state
- saves: player_id, world_id, save_blob, last_played
- world_places: id, map_id, coordinates, category
- world_inventory: player, item, quantity, obtained_at
- npc_memory: NPC, player, memory_type, event_id, valence
- world_schedule: NPC, date_range, place, plan
- arcade_sessions: game, player, score, outcome, occurred_at

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

COMMERCE_LEGAL TABLE FAMILY

- content_rights: work/asset, proof, owner, licenses, start/end
- assets: id, blob_id, attribution, alt_text, content_rating
- orders_mock: player, store, item, status, paid_virtual
- event_tickets_mock, inventory_stock, store_hours
- audit_logs: actor, action, subject, timestamp, IP hash
- consent_records: user, scope, granted_at, revoked_at

Family invariants

IDs are stable UUIDs. Relationships use foreign keys. Times are timezone-aware and convert from source-local calendar values explicitly. Visibility is checked in the API, not only hidden in the frontend. Add indexes for common joins, source provenance, publication dates and user ownership.

Migration and seed strategy

- 001 identity and roles; 002 provenance+canon; 003 franchise/media; 004 social; 005 desktop; 006 games/world; 007 legal/moderation; 008 historical temporal indexes.
- A deterministic 2012 seed loads Greg and friends, YAL sample boards, MeTube channel IDs, and offline place map placeholders.
- A deterministic 2013 seed loads IoE/ITR premiere records, initial 3 discussion pages and time-tested counts.
- Seed runner must be idempotent: running twice creates no duplicate personas, releases or forum posts.
- Separate production data migrations from content bundles, allowing copyright review and future canonical updates.

CHAPTER 22 / MASTER REFERENCE

API and cross-application contract

Stable APIs allow the browser, websites and game worlds to exchange information without duplicating application logic.

Verb	Endpoint	Behavior
GET	/v1/history/snapshot?at=...	Dated content manifest and available services
GET	/v1/search?q=...&at=...	Period-filtered ranked results
GET	/v1/sites/{site}/pages/{slug}	Resolved functional fictional webpage
GET	/v1/anime?season=fall-2013	Filtered anime catalogue
GET	/v1/anime/{id}	Anime details, credits and links
GET	/v1/anime/{id}/episodes	Episodes available by date
GET	/v1/manga/{id}/chapters	Readable chapters and rights
GET	/v1/manga/chapters/{id}/pages	Secure page token manifest
PUT	/v1/users/me/reading-state	Update manga progress
PUT	/v1/users/me/anime-list/{id}	Set status/score/progress
GET	/v1/threads/{id}?page=...	Forum with archived and modern overlays
POST	/v1/threads/{id}/posts	Write a modern-authored reply
POST	/v1/reports	Submit a moderation report
GET	/v1/videos/{id}	Video detail and renditions
POST	/v1/videos/uploads	Initiate authenticated upload
POST	/v1/videos/{id}/comments	Authenticated comment
PUT	/v1/users/me/subscriptions/{id}	Subscribe or unsubscribe
GET	/v1/me/messages	Private conversation threads
POST	/v1/me/messages	Send authorized message
GET	/v1/me/desktop/session	Restore window and app state
PUT	/v1/me/desktop/session	Persist current workspace
POST	/v1/me/downloads	Create validated download job
GET	/v1/me/files?folder=...	List virtual directory
POST	/v1/me/files/move	Atomic rename/move
PUT	/v1/me/preferences	Update settings and theme
GET	/v1/world/{id}/places	World map and access state
GET	/v1/world/{id}/npcs?at=...	NPC locations and dialogue branch
POST	/v1/world/{id}/actions	Apply game event and save delta

Verb	Endpoint	Behavior
GET	/v1/stores/{id}/stock?at=...	Period-specific physical stock
GET	/v1/music/stations/{id}/schedule	Radio program and track rights
GET	/v1/assets/{id}	Metadata and authorized signed URL
GET	/v1/canon/decisions	Read editorial decision provenance

Common API rules

- Version all endpoints and payloads.
- Require an authenticated modern identity for writes; fictional persona archives are immutable by ordinary members.
- Every response includes origin, context date, actual request time and visibility rule when applicable.
- Use cursor pagination for evolving feeds; page numbers are acceptable for classic YAL forum presentation.
- Rate-limit expensive search, registration, upload and write endpoints.
- Expose typed structured errors for no-rights, missing content, not-yet-published, network offline and service unavailable.

Core event model

- anime.episode_aired: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- article.published: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- forum.thread_created: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- forum.post_published: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- video.published: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- file.download_completed: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- user.bookmark_saved: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- world.inventory_added: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- world.npc_memory_changed: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- store.stock_changed: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.
- music.track_released: event_id, subject_id, actor, occurred_at, published_at, origin, payload_schema_version.

CHAPTER 23 / MASTER REFERENCE

Safety, privacy, rights and content governance

The early internet can look unruly without recreating its dangerous technical vulnerabilities.

Mandatory modern security

- Electron: context isolation, sandboxing, no Node.js integration for remote pages; permission restrictions and navigation allowlists [W07].
- Backend: every sensitive read/write authenticated and authorization tested. Enable PostgreSQL row-level security where exposed [W12].
- Files: extension allowlist, signature sniff, caps, filename normalization, malware scans, quarantine, separate storage from application execution [W13].
- Sensitive accounts: short-lived access tokens, secure cookie settings, CSRF protections, passkeys/MFA where practical.
- Moderator actions logged; privacy export and delete mechanisms; retention schedule explicit.
- Secrets only in environment managers; rotate leaked credentials; prevent them from appearing in seed databases, logs or generated docs.

IP, copyright and brand strategy

Lumine original manga, animation, music, artwork and fictional platforms can be fully implemented subject to project ownership. Third-party entertainment, brands, exact logos, screenshots and commercial media may require permission or limited quotation/reference use. Favor clearly inspired original browser and search branding unless legal review authorizes actual marks and UI assets. Never market a reconstructed branded service as official or claim it provides current historical Google infrastructure.

User-generated content

- Copyright takedown path; uploader attests rights; strike/appeal workflow.
- Age-appropriate content ratings for public media; private accounts are not automatically public.
- Abuse reports for spam, harassment and illegal material; rate limits; mod controls.
- Spam-resistant sign-up, bot detection and comment throttling.
- Optional content filters for images, manga and media uploads.

Historical archive ethics

Historical-looking accounts are reconstructions. Public documentation must label them as authored data, and archived user interactions must be distinguishable from current live conversations. Do not imitate real private individuals or create fabricated evidence about real people.

Accessibility

Target WCAG 2.2 AA for modern implementation: keyboard navigation, focus indicators, text alternatives, accessible forms, captions, contrast, reduced motion and zoom [W14]. Period styling must not undermine accessibility.

CHAPTER 24 / MASTER REFERENCE

Performance, offline mode, observability and deployment

Engineering operations are part of a functioning product, not afterthoughts.

Performance budgets (initial targets; validate under test)

Surface	Target	Measurement
Static historical pages	LCP <= 2.5s on typical broadband test	Synthetic browser tests
Search	p95 API <= 500ms for initial index	Tracing and load tests
Desktop app launch	Interactive shell <= 5s on reference midrange PC	Cold-start benchmark
Local browsing navigation	p95 <= 250ms for cached internal pages	Instrumentation
Forum paging	<= 1.5s with 20 posts/page	Playwright
Video playback	<= 4s to first frame on test clip	Player telemetry
Download persistence	Verified after restart and clean shutdown	Integration test

Offline operation

- Service worker for installable web option; cache first-party historical page shells and permitted offline assets [W15].
- Local files stored in OPFS/IndexedDB or desktop private storage; quotas and eviction explicitly surfaced to the user [W08, W09].
- Read-only offline content can be accessed while network is unavailable.
- New writes queue with idempotency keys; sync conflict UI, no silent overwriting of forum posts.
- Download jobs clearly distinguish remote/active/cached/completed and file availability.

Observability

- Collect structured logs for errors and service health without exposing message bodies, passwords or private files.
- Monitoring: request latency, queue delays, failed uploads, search misses, media errors, archive gate violations.
- Audit log source edits, moderator actions, content rights updates and release schedule changes.
- Weekly restore tests from backups; automated migration validation before deploy.

Release plan

Local developer environment first, internal test deployment second, opt-in beta third, then public release if content rights, moderation staffing, infrastructure budget and support are in place. Never assume user accounts or servers already exist merely because prototypes were generated.

CHAPTER 25 / MASTER REFERENCE

End-to-end user journeys and cross-platform integration

Each journey has an objective, required operations, persistent output and verifiable completion conditions.

J-01 Greg boots his PC

User path: Launch desktop, load account, restore wallpaper and Messages. Expected outcome: Restores saved layout + read state.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-02 Search to YAL

User path: Search Impact of Eden, open YAL result, add to Watchlist. Expected outcome: Bookmark and list entry share anime ID.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-03 YAL to Episode 1 forum

User path: Browse Oct 6 2013 and open thread after 02:31 JST. Expected outcome: No pre-thread posts before creation.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-04 YAL to MeTube

User path: From IoE related media open creator video in a new tab. Expected outcome: Both page links resolve correctly.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-05 MeTube to download

User path: Download authorized wallpaper; file moves to virtual Downloads. Expected outcome: Actual blob matches Explorer metadata.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-06 Download to desktop wallpaper

User path: Open downloaded image, select Set Wallpaper, reboot. Expected outcome: Chosen wallpaper survives restart.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-07 Messenger shares content

User path: Send YAL page URL to Lexi; click URL in chat. Expected outcome: URL and sender metadata retained.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-08 Manga borrow and read

User path: Receive Volume 1 in Episode 2; read page subset at night. Expected outcome: Persistent book and progress affect dialogue.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-09 Playlist during browsing

User path: Play MP3 and open another browser tab. Expected outcome: Audio continues under window changes.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-10 Arcade score callback

User path: Play racing cabinet; later chat mentions outcome. Expected outcome: World event links to NPC memory.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-11 Retail stock

User path: Browse bookstore on Oct 5 2013; find available manga and magazines. Expected outcome: Availability dates restrict inventory.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-12 Historical search gate

User path: Search Oct 5 for ep1 YAL reply made Oct 7. Expected outcome: Result is absent.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-13 Future anime gate

User path: Browse 2012 YAL while IoE TV has yet to premiere. Expected outcome: Manga and older OVAs can exist, 2013 TV not yet aired.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-14 Review moderation

User path: Report abusive modern comment, mod hides and can restore. Expected outcome: User and audit states consistent.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-15 Web external blocked

User path: Open site refusing iframe, offer system browser choice. Expected outcome: No insecure bypass.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-16 Video rights change

User path: Media license expires and playback is disabled. Expected outcome: Historical metadata remains readable.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-17 Old username provenance

User path: View YALBot and older usernames. Expected outcome: Reconstructed origin is inspectable.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-18 Switch device profile

User path: Different user logs in and opens PC. Expected outcome: Private files isolated.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-19 Internet disconnected

User path: Browse cached local manga and files without network. Expected outcome: No data loss or deceptive connectivity.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-20 Upgrade era skin

User path: Move to a later dated UI, browse old page. Expected outcome: Original 2013 archival view retained in replay.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-21 Export personal data

User path: Download list, saves, posts, preferences and files. Expected outcome: Portable export succeeds.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-22 Data restore

User path: Restore an old save; verify no date-locked content leaked. Expected outcome: Time and source integrity conserved.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-23 Cross-service live conversation

User path: Use modern comment on a 2013 archived page. Expected outcome: Modern timestamp and overlay clearly marked.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

J-24 Search old studio staff

User path: Find Keito Kisaragi and open earlier Flow credits. Expected outcome: Career edges cross studios without ownership confusion.

- Must exercise at least two distinct application modules or a stateful module and desktop service.
- Verify both UI behavior and authoritative database state; log errors with reproducible route and timestamp.

CHAPTER 26 / MASTER REFERENCE

QA acceptance matrix and release gates

The product is complete only when objective checks pass. Visual resemblance alone never counts as function.

Identity

- IDE-01: fresh registration - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- IDE-02: sign in/out - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- IDE-03: expired session - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- IDE-04: password reset - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- IDE-05: duplicate user - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Security

- SEC-01: SQL/XSS sanitization - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SEC-02: CSRF - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SEC-03: RLS - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SEC-04: unauthorized file read - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SEC-05: malicious protocol - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Desktop

- DES-01: window focus - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- DES-02: minimize - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- DES-03: drag - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- DES-04: maximize - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- DES-05: restore - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- DES-06: restart - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- DES-07: two profiles - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Files

- FIL-01: CRUD folders - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- FIL-02: duplicate - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- FIL-03: Trash restore - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- FIL-04: filename sanitation - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- FIL-05: quota - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Browser

- BRO-01: tabs - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- BRO-02: history - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- BRO-03: bookmarks - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- BRO-04: 404 - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- BRO-05: certificate - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- BRO-06: live page blocked - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Search

- SEA-01: phrase - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SEA-02: fuzzy - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SEA-03: filters - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SEA-04: date gating - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SEA-05: multilingual titles - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

YAL Anime

- YAL-01: season grid - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- YAL-02: studio/people pages - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- YAL-03: edit list - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- YAL-04: score stats - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

YAL Forum

- YAL-01: quote - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- YAL-02: spoiler - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- YAL-03: pagination - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- YAL-04: report - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- YAL-05: created-at constraint - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

MeTube

- MET-01: upload/transcode - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MET-02: stream - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MET-03: seek - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MET-04: captions - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MET-05: subscribe - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MET-06: comment - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Manga

- MAN-01: RTL/LTR - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MAN-02: page caching - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MAN-03: bookmark - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MAN-04: license block - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MAN-05: resume - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Messaging

- MES-01: buddy state - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MES-02: chat - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MES-03: attachment - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MES-04: unread - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MES-05: moderation - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Social

- SOC-01: post - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SOC-02: edit - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SOC-03: repost - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SOC-04: follow - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SOC-05: image upload - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- SOC-06: reports - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Music

- MUS-01: playlist - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MUS-02: seek - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MUS-03: shuffle - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MUS-04: local file association - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- MUS-05: radio schedule - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Games

- GAM-01: launch - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- GAM-02: input - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- GAM-03: score - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- GAM-04: save - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- GAM-05: sandbox - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- GAM-06: compatibility error - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Offline Worlds

- OFF-01: NPC routes - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- OFF-02: July 12 branch - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- OFF-03: book - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- OFF-04: arcade - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

- OFF-05: message callbacks - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Historical clock

- HIS-01: time-zone DST - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- HIS-02: unpublished item - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- HIS-03: captured metric - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- HIS-04: valid range - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Content ledger

- CON-01: superseded names - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- CON-02: conflicting versions - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- CON-03: source and canon edits - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Legal

- LEG-01: provenance - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- LEG-02: right-holder proof - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- LEG-03: takedown - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- LEG-04: age rating - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Accessibility

- ACC-01: keyboard - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- ACC-02: contrast - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- ACC-03: captions - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- ACC-04: screen reader - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- ACC-05: focus - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Performance

- PER-01: LCP - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- PER-02: API p95 - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.
- PER-03: media time-to-first-frame - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

- PER-04: backup restore - PASS requires successful UI action, verified persistent state, recorded automated test, and failure-path coverage.

Release gates

Gate	Required evidence
R0 Design verified	All feature contracts tagged and referenced, initial screens reviewed.
R1 Functional desktop	File CRUD, apps and boot persistence tests passing.
R2 Historical web	Search/YAL/MeTube/reader/communications usable in 2012-13.
R3 Integration	Ten core user journeys with cross-app persistence.
R4 Security/content	Threat model, upload scans, moderation, legal clearance and accessibility.
R5 Pilot	Real users complete the Oct 5-6 2013 scenario without manual intervention.
R6 2014+ expansion	Only after R5 and content review; optional features isolated.

CHAPTER 27 / MASTER REFERENCE

Implementation roadmap with dependency order

Executable work packages with owner competencies, deliverables, testing and finish lines. This is a plan, not a claim that the site is already built.

Package	Planning range	Deliverables	Definition of done
P0 - Source lock and assets	2-4 design sprints	Source registry, canon decisions, asset licenses, seeded October 2013 dataset.	Every shipped datum has provenance; 2012/13 constraints tested.
P1 - Shared back end	3-5 engineering sprints	Postgres schema, auth, search, event store, storage, REST API, migrations.	API contract tests and permissions pass.
P2 - Connected Desktop	3-5 engineering sprints	Windows/OS shell, file explorer, downloads, browser frame, settings.	Reboot workflow, app window tests and file state pass.
P3 - YAL 2013 vertical slice	4-6 engineering sprints	Anime details, three IoE episode forums, lists, profiles, genre/industry search.	Screenshot QA, historical visibility, real forum actions.
P4 - Video, manga, media	4-6 engineering sprints	MeTube upload/player/channel, original manga reader, music/radio, authorized downloads.	Media flows run with licensed content and stable file state.
P5 - Communication, social, games	4-6 engineering sprints	Messenger, Birdie/Tum/forum, basic games and original Flash-style mini experiences.	Cross-service, moderation and save workflows pass.
P6 - OFFLINE connection	3-6 engineering sprints	Greg home PC, July 11-12 scene hooks, mall/bookstore/arcade integration.	Reproducible two-day playable scenario.
P7 - Beta / operations	2-4 hardening sprints	Accessibility, security, crash analytics, backup, legal review, pilot.	R5 sign-off on exact hardware baseline.
P8 - Subsequent years	Continuous after beta	Release year packets 2014 through 2026, studio catalog expansion.	Each year independently passes no-future-leak tests.

Sprint ranges are comparative sizing estimates only. Actual delivery dates depend on engineering team, content production rights and infrastructure. The initial objective is a single fully working 2012-2013 experience, not 21 simultaneously incomplete years.

Roles needed

- Product owner / canon lead: signs creative conflicts and release policy.
- Technical lead: architecture, data model, application security, deployment.
- Frontend/desktop engineer: window manager and site chrome.
- Backend/search engineer: database, query semantics, media upload and APIs.
- Game engineer/world designer: OFFLINE town and NPC persistence.
- Historical researcher/editor: date, technology, citations and provenance.
- Artist/audio and licensing manager: permitted assets, user content policy, music rights.
- QA/accessibility: keyboard, performance, browser versions and scripted end-to-end scenarios.

CHAPTER 28 / MASTER REFERENCE

Open canon and business decisions: controlled register

Remaining choices do not invalidate the 100% functional specification. Each has a safe placeholder and an explicit consequence.

ID	Issue	Choice / conflict	Interim binding rule	Owner
CAN-001	OFFLINE town name	Unnamed vs Crescent Valley vs Branford	Use world_id and “Town” display name until approved.	Project owner
CAN-002	Alternate friend group	Lucas/Mia/Jordan/Erin vs Greg/Ren/Jay/Lexi	Use Greg group; keep alternate draft isolated.	Canon editor
CAN-003	Greg family	Mom name, father existence, ages/last names	Use “Mom”, no ungrounded family addition.	Story editor
CAN-004	IoE total episode count	206 vs 226 archived master variants	Version-indexed totals; only confirmed initial run public.	Franchise producer
CAN-005	IoE OVA legacy data	Nightpolis cast, original studio, episode count	Pending records, not invented.	Research editor
CAN-006	ITR sequel precise dates	S2P1/S3/S4 working schedules	Label as working; keep S2P2 approved.	Producer
CAN-007	ITR sequel animation	Studio credits for Seasons 2-4	Unset until verified; never assign CHAOS Home blindly.	Production editor
CAN-008	Independent studios	30-50 target vs 4 established + proposals	Proposed studios blocked from historical browse until approved.	Studio directory owner
CAN-009	Studio Burst catalogue detail	Staff, release dates for supporting titles	Show documented season/format only.	Production editor
CAN-010	Starfall & ARATAKI	Exact releases/credits and out-of-range 2027 plan	Draft records; ARATAKI outside 2006-26 main filter.	Franchise producer
CAN-011	Soniplex vs AniBlex	Company relationship unresolved	Keep separate entities with possible relation.	Industry editor
CAN-012	IoE fiction within Greg world	Mirror/prophecy/red herring relationship	Do not script reveal or force reader narrative.	Story producer
CAN-013	OFFLINE scope	How much driving, jobs, weather playable at launch	Prototype subset then gate expand.	Game lead
CAN-014	Real brand visual strategy	Actual Google/Firefox vs inspired fiction	Use inspired UI and external link option until legal review.	Legal/product
CAN-015	Monetization	Subscriptions, real purchases, free access	No real-money payments in first slice.	Business lead
CAN-016	Public UGC launch	Open accounts vs closed beta	Private pilot with moderation first.	Trust/Safety

ID	Issue	Choice / conflict	Interim binding rule	Owner
CAN-017	Content licensing	Commercial video/manga/music rights	Original or explicitly cleared assets only.	Rights owner
CAN-018	Historical discovery scale	All 21 years vs first 2-year slice	Build 2012-13 first; dormant schema covers later.	Product owner
CAN-019	Media source version conflicts	Multiple versions of older archived files	Provenance chain and source precedence; no overwrite.	Editorial archivist
CAN-020	Screen artwork	How exact screenshots are used	Reference composition only; recreate with own assets.	Design lead

No invented approvals

An unresolved canon choice is not a missing engineering requirement. The product can be fully architected with null-safe data, source fields and editorial tooling while preserving a transparent decision gate.

CHAPTER 29 / MASTER REFERENCE

Source catalogue and research references

External technical facts are grounded in official product documentation, standards, archival services and research institutions.

W01 Pew Research Center: Social Media Usage: 2005-2015

https://www.pewresearch.org/internet/2015/10/08/social-networking-usage-2005-2015/

W02 Microsoft: Windows 7 lifecycle and general availability

https://learn.microsoft.com/en-us/lifecycle/products/windows-7

W03 Microsoft: Windows 8 Arrives, Oct 26 2012 launch

https://news.microsoft.com/source/2012/10/25/windows-8-arrives/

W04 Mozilla MDN: Firefox 24 release Sept 17 2013

https://developer.mozilla.org/en-US/docs/Mozilla/Firefox/Releases/24

W05 Mozilla Firefox: Firefox 29 release Apr 29 2014

https://www.firefox.com/en-US/firefox/29.0/releasenotes/

W06 Google Chrome Releases: Chrome 30 Stable Oct 1 2013

https://chromereleases.googleblog.com/2013/10/stable-channel-update.html

W07 Electron: Electron Security Recommendations

https://www.electronjs.org/docs/latest/tutorial/security

W08 MDN: Origin private file system

https://developer.mozilla.org/en-US/docs/Web/API/File_System_API/Origin_private_file_system

W09 MDN: IndexedDB API

https://developer.mozilla.org/en-US/docs/Web/API/IndexedDB_API

W10 Internet Archive: Using The Wayback Machine - limitations

https://archivesupport.zendesk.com/hc/en-us/articles/360004651732-Using-The-Wayback-Machine

W11 Ruffle: Flash emulator project / compatibility

https://github.com/ruffle-rs/ruffle

W12 Supabase: PostgreSQL Row-Level Security

https://supabase.com/docs/guides/database/postgres/row-level-security

W13 OWASP: File Upload Cheat Sheet

https://cheatsheetseries.owasp.org/cheatsheets/File_Upload_Cheat_Sheet.html

W14 W3C: WCAG 2.2 Recommendation

https://www.w3.org/TR/wcag/

W15 MDN: Progressive Web Apps offline operation

https://developer.mozilla.org/en-US/docs/Web/Progressive_web_apps

W16 Electron: WebContentsView API

https://www.electronjs.org/docs/latest/api/web-contents-view

W17 MDN: CSP frame-ancestors

https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/Content-Security-Policy/frame-ancestors

W18 Pew Research: Smartphone Ownership 2013

https://www.pewresearch.org/internet/2013/06/05/smartphone-ownership-2013/

W19 Pew Research: Mobile Device Ownership time series

https://www.pewresearch.org/internet/fact-sheet/mobile/

W20 MDN: File System API permissions

https://developer.mozilla.org/en-US/docs/Web/API/File_System_API

Primary user-provided / prior-work source documents

ID	Document	Used for
F01	Offline Offline World Bible.pdf	Episode 1, July 2012 cast, PC, messaging, radio, town ambiguity.
F02	Impact_of_Eden_Offline_Mode_Episode_2_Planning.pdf	Full July 12 2012 planning, persistence, mall, arcade and reading.
F03	yal_impact_of_eden_ep1_2013.html, ep2_2013.html, ep3_2013.html	Primary YAL screenshots/HTML discussion styling, dates, users, counts.
F04	yal_impact_of_eden_nightpolis_winter_2005_YAL_matched.html and MAL_style.html	Historic OVA page structure and open credits.
F05	Iron_to_Requiem_S1_Episode_Guide_Axel_Updated(1).docx	Season 1 26 Thursday broadcasts and production identity.
F06	Studio%20Burst%20Identity.pdf.pdf	20-title Burst slate and Moonphase Riot five-season record.
F07	LUNE_Flow_Pictures_Studio_Burst_Dossier_2026.pdf	LUNE independent career, song and producer relationships.
F08	Flow_Pictures_Archive_Reedited(3).pdf	Early films, television catalogue, staff employment and historical credits.
F09	Iron_to_Requiem_Season_1_Lore_Updated(1).pdf	Older four-season structure 87 eps, now superseded by S2P2 update.
F10	EARTH: Metal Comet anime air-date PDF records	Multi-season broadcast dates and studios (multiple documents).
F11	Impact of Eden TV broadcast master and continuous-index updates	Premiere master and conflicting later indexed totals.
F12	Alive vs Dead Internet Theory(20261024-210902).pdf	Period culture vocabulary, online/offline evolution interpretive source.

Note: Some file names are literal library names; all source-derived claims are summarized and interpreted. This document was authored from accessible excerpts and selected full-text reads. It is not a guarantee that every file ever created in Lumine StarWorks has been audited.

CHAPTER 30 / MASTER REFERENCE

Completion criteria and handoff

Specification coverage is complete for Edition 1.0. Delivery of functioning applications is a separate future implementation milestone.

What this document now covers

- Complete product taxonomy from 2006 through 2026, with initial build priority anchored 2012-13.
- Historical source/provenance rules, date visibility and publication/capture separation.
- Usable desktop, operating-system shell, browser, search, file manager, video, manga, anime, chat, social, games, music and offline venues.
- Full reference registers for the documented Burst and Flow titles plus major franchises and soundtrack lineage.
- Shared relational schema, API contracts, cross-app workflow IDs, security and licensing requirements.
- Explicit testing matrix, release gates, reference sources and canon decision register.

What the team still has to build

Implement the software, create/clear media assets, finalize open franchise canon facts, run moderation, provision servers, test hardware, and complete all acceptance gates. None of these are claimed done by writing this specification. The earlier Claude Desktop screenshot showed model-provider authentication problems; no evidence from that screenshot proves deployment.

Definition of document completion

Edition 1.0 readiness
100% of the intended master-specification chapters are authored and interconnected; the project itself is not 100% implemented, and open story/content decisions remain transparently registered. This is the correct boundary between a finished design bible and a finished live product.

Immediate next work package

Resolve Claude Code authentication safely, without sharing credentials. Create a project repository and implement the initial 2012-13 vertical slice: Connected Desktop + Browser + Search + YAL Episode 1-3 + MeTube playback + Reader + Messages + Greg persistence. After it passes the acceptance journeys, extend the content era-by-era. No jump to 2016 is required for the initial YAL delivery.

CHAPTER 31 / MASTER REFERENCE

APPENDIX A - comprehensive application requirements

This appendix converts every major service into explicit interaction requirements. Rows are commitments for implementation, not evidence that the applications are already live.

Connected Desktop - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
CON-01	Power-on and user picker	Persist + time-gate + test
CON-02	Lock/log out/reboot/shut down	Persist + time-gate + test
CON-03	Start menu and program launcher	Persist + time-gate + test
CON-04	Window focus, drag, resize, minimize	Persist + time-gate + test
CON-05	Taskbar pins and running application indicators	Persist + time-gate + test
CON-06	Virtual folders and file CRUD	Persist + time-gate + test
CON-07	Drag and drop file objects across apps	Persist + time-gate + test
CON-08	Files and app associations	Persist + time-gate + test
CON-09	Downloads progress and retry	Persist + time-gate + test
CON-10	Notification center and sound controls	Persist + time-gate + test
CON-11	Theme and wallpaper changes	Persist + time-gate + test
CON-12	Multi-user partition and save restore	Persist + time-gate + test
CON-13	Network status and offline mode	Persist + time-gate + test
CON-14	Local device time vs historical time badge	Persist + time-gate + test
CON-15	Accessibility settings and zoom	Persist + time-gate + test

Browser and search - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
BRO-01	Enter fictional and external URL	Persist + time-gate + test
BRO-02	Tab history and closed tab restore	Persist + time-gate + test
BRO-03	Back, forward, reload and home	Persist + time-gate + test
BRO-04	Bookmarks and folders	Persist + time-gate + test

Req ID	Function or capability	Verification
BRO-05	Browsing history search and wipe	Persist + time-gate + test
BRO-06	Downloads and mime routing	Persist + time-gate + test
BRO-07	Find in page and zoom	Persist + time-gate + test
BRO-08	Dated content query filter	Persist + time-gate + test
BRO-09	Search images and video result types	Persist + time-gate + test
BRO-10	Autocomplete and fuzzy search	Persist + time-gate + test
BRO-11	Privacy mode isolated storage	Persist + time-gate + test
BRO-12	User confirmation for external sites	Persist + time-gate + test
BRO-13	Blocked frame fallback to external browser	Persist + time-gate + test
BRO-14	Website certificate and protocol warning	Persist + time-gate + test
BRO-15	Session restore after crash	Persist + time-gate + test

YAL - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
YAL-01	Anime directory and advanced filters	Persist + time-gate + test
YAL-02	Anime profile and broadcast information	Persist + time-gate + test
YAL-03	Manga directory and reader links	Persist + time-gate + test
YAL-04	Franchise release relationship graph	Persist + time-gate + test
YAL-05	Characters and favorite toggles	Persist + time-gate + test
YAL-06	Staff studio and composer linkages	Persist + time-gate + test
YAL-07	Member registration and profiles	Persist + time-gate + test
YAL-08	Personal anime list status and progress	Persist + time-gate + test
YAL-09	Reviews and recommendation submissions	Persist + time-gate + test
YAL-10	Discussion thread paging	Persist + time-gate + test
YAL-11	Quote, spoiler and BBCode	Persist + time-gate + test
YAL-12	Historical captured score snapshots	Persist + time-gate + test
YAL-13	Seeding Episode 1-3 old posts	Persist + time-gate + test
YAL-14	Create and follow modern discussions	Persist + time-gate + test
YAL-15	User report, moderator lock and audit	Persist + time-gate + test

MeTube - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
MET-01	Video upload, transcode and preview	Persist + time-gate + test
MET-02	Publish, edit and remove video	Persist + time-gate + test
MET-03	Creator channel and profile	Persist + time-gate + test
MET-04	Video playback and buffering states	Persist + time-gate + test
MET-05	Seek, quality, speed and subtitles	Persist + time-gate + test
MET-06	Watch later and playlists	Persist + time-gate + test
MET-07	Search and recommendation results	Persist + time-gate + test
MET-08	Comment and quote reply	Persist + time-gate + test
MET-09	Like/favorite and subscribe	Persist + time-gate + test
MET-10	Creator activity dashboard	Persist + time-gate + test
MET-11	Video share link and embed policy	Persist + time-gate + test
MET-12	Copyright report and takedown	Persist + time-gate + test
MET-13	Thumbnails and caption generation	Persist + time-gate + test
MET-14	Capture historic view snapshots	Persist + time-gate + test
MET-15	Store local authorized video download	Persist + time-gate + test

Manga reader - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
MAN-01	Series and volume listing	Persist + time-gate + test
MAN-02	Chapter page manifest	Persist + time-gate + test
MAN-03	Right-to-left mode	Persist + time-gate + test
MAN-04	Two-page spread and single-page mode	Persist + time-gate + test
MAN-05	Zoom, pan and fit width	Persist + time-gate + test
MAN-06	Reading position resume	Persist + time-gate + test
MAN-07	Page/chapter bookmarks	Persist + time-gate + test
MAN-08	Offline cached authorized chapters	Persist + time-gate + test
MAN-09	Chapter release timestamp gating	Persist + time-gate + test
MAN-10	Member scores and notes	Persist + time-gate + test
MAN-11	Separate preview and licensed full read	Persist + time-gate + test

Req ID	Function or capability	Verification
MAN-12	Uploader rights and access constraints	Persist + time-gate + test
MAN-13	Chapter page load error and retry	Persist + time-gate + test
MAN-14	Image credits and attribution	Persist + time-gate + test
MAN-15	Import private CBZ/PDF/images by permission	Persist + time-gate + test

Communication - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
COM-01	Buddy contact list	Persist + time-gate + test
COM-02	Away/online/offline states	Persist + time-gate + test
COM-03	Private messaging	Persist + time-gate + test
COM-04	Group messaging and history	Persist + time-gate + test
COM-05	Notification and unread counts	Persist + time-gate + test
COM-06	Text attachment shares	Persist + time-gate + test
COM-07	Image attachment preview	Persist + time-gate + test
COM-08	Message search and export	Persist + time-gate + test
COM-09	Block contact and report	Persist + time-gate + test
COM-10	Email inbox and composing	Persist + time-gate + test
COM-11	Email search and drafts	Persist + time-gate + test
COM-12	Email reply/forward and threading	Persist + time-gate + test
COM-13	Offline queue of pending messages	Persist + time-gate + test
COM-14	Distinguish historical NPC and live messages	Persist + time-gate + test
COM-15	Consent and data deletion	Persist + time-gate + test

Social websites - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
SOC-01	Birdie period character limit	Persist + time-gate + test
SOC-02	Birdie followers and mentions	Persist + time-gate + test
SOC-03	Birdie hashtags and timelines	Persist + time-gate + test
SOC-04	Tum themeable profile	Persist + time-gate + test
SOC-05	Tum reblog and artwork sets	Persist + time-gate + test

Req ID	Function or capability	Verification
SOC-06	3Chan anonymous board structure	Persist + time-gate + test
SOC-07	3Chan thread expiration rules	Persist + time-gate + test
SOC-08	Forum BBCode/quoting and signatures	Persist + time-gate + test
SOC-09	RSS feed and subscriptions	Persist + time-gate + test
SOC-10	User personal website builder	Persist + time-gate + test
SOC-11	Fan fiction chapter and comment	Persist + time-gate + test
SOC-12	Fan art authorized gallery	Persist + time-gate + test
SOC-13	Volunteer moderator rights	Persist + time-gate + test
SOC-14	Spam throttling and abuse report	Persist + time-gate + test
SOC-15	Period-aware feed sorting	Persist + time-gate + test

Games & launcher - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
GAM-01	Original playable HTML5 games	Persist + time-gate + test
GAM-02	Ruffle-compatible SWF support	Persist + time-gate + test
GAM-03	Hardware keyboard/gamepad input	Persist + time-gate + test
GAM-04	Save state and autosave	Persist + time-gate + test
GAM-05	Leaderboard with historical partition	Persist + time-gate + test
GAM-06	Game catalogue and release metadata	Persist + time-gate + test
GAM-07	Player-owned screenshots	Persist + time-gate + test
GAM-08	MeTube gameplay share	Persist + time-gate + test
GAM-09	Game-specific forum page	Persist + time-gate + test
GAM-10	Authorized downloadable game files	Persist + time-gate + test
GAM-11	Arcade credits and score persistence	Persist + time-gate + test
GAM-12	In-world NPC callback	Persist + time-gate + test
GAM-13	Controls remapping	Persist + time-gate + test
GAM-14	Accessibility and pause	Persist + time-gate + test
GAM-15	Secure isolation of untrusted game content	Persist + time-gate + test

Audio and radio - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
AUD-01	Local import of authorized audio	Persist + time-gate + test
AUD-02	MP3/OGG playback	Persist + time-gate + test
AUD-03	Track metadata and album art	Persist + time-gate + test
AUD-04	Playlist and queue editing	Persist + time-gate + test
AUD-05	Shuffle/repeat	Persist + time-gate + test
AUD-06	Seek and volume mix	Persist + time-gate + test
AUD-07	Background playback across windows	Persist + time-gate + test
AUD-08	Radio station schedule	Persist + time-gate + test
AUD-09	Program/advert IDs and date gating	Persist + time-gate + test
AUD-10	Track license and expiration	Persist + time-gate + test
AUD-11	Lyrics only when authorized	Persist + time-gate + test
AUD-12	Music releases in search	Persist + time-gate + test
AUD-13	Original event DJ announcements	Persist + time-gate + test
AUD-14	Saved favorites	Persist + time-gate + test
AUD-15	Error state for missing local file	Persist + time-gate + test

Offline town - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
OFF-01	Home and room interactions	Persist + time-gate + test
OFF-02	Street traversal and discoverable places	Persist + time-gate + test
OFF-03	Weather and daily time	Persist + time-gate + test
OFF-04	NPC position and schedule	Persist + time-gate + test
OFF-05	Inventory and money	Persist + time-gate + test
OFF-06	Book loan and reading progression	Persist + time-gate + test
OFF-07	Mall, bookstore and food court	Persist + time-gate + test
OFF-08	Arcade mini-games	Persist + time-gate + test
OFF-09	Friend-specific optional branches	Persist + time-gate + test
OFF-10	Group-chat photo memory	Persist + time-gate + test
OFF-11	Family conversations	Persist + time-gate + test
OFF-12	Physical media shop inventory	Persist + time-gate + test
OFF-13	Radio/music in vehicles or stores	Persist + time-gate + test
OFF-14	Day-end saving and state restore	Persist + time-gate + test

Req ID	Function or capability	Verification
OFF-15	Out-of-period posters blocked	Persist + time-gate + test

Industry archives - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
IND-01	Studio page and founding date	Persist + time-gate + test
IND-02	Staff credit relations	Persist + time-gate + test
IND-03	Producer vs licensor separation	Persist + time-gate + test
IND-04	Original and alternate title records	Persist + time-gate + test
IND-05	Anime episode release schedules	Persist + time-gate + test
IND-06	Movie and OVA entries	Persist + time-gate + test
IND-07	Soundtrack release catalog	Persist + time-gate + test
IND-08	Physical merchandise registry	Persist + time-gate + test
IND-09	Convention and screening events	Persist + time-gate + test
IND-10	Regional broadcast licenses	Persist + time-gate + test
IND-11	Original trailer metadata	Persist + time-gate + test
IND-12	Correction and revision audit	Persist + time-gate + test
IND-13	Title and person aliases	Persist + time-gate + test
IND-14	Multiple documentary sources	Persist + time-gate + test
IND-15	Open canon issue references	Persist + time-gate + test

Admin and platform - acceptance contract

Each item below is a user- or editor-visible operation. Acceptance requires a real state transition, permission check, error state and reproducible regression test.

Req ID	Function or capability	Verification
ADM-01	Source import with revision hash	Persist + time-gate + test
ADM-02	Canon approval workflow	Persist + time-gate + test
ADM-03	Moderation and appeals	Persist + time-gate + test
ADM-04	Asset license tracking	Persist + time-gate + test
ADM-05	Storage quota and budget alerts	Persist + time-gate + test
ADM-06	System health and log viewer	Persist + time-gate + test
ADM-07	Backups and restoration	Persist + time-gate + test
ADM-08	Regional rights flags	Persist + time-gate + test

Req ID	Function or capability	Verification
ADM-09	Content/timeline seed rollback	Persist + time-gate + test
ADM-10	Public transparency labels	Persist + time-gate + test
ADM-11	User data export and delete	Persist + time-gate + test
ADM-12	Automated regression dashboard	Persist + time-gate + test
ADM-13	Performance load testing	Persist + time-gate + test
ADM-14	Access control audits	Persist + time-gate + test
ADM-15	Feature flags and staged releases	Persist + time-gate + test

CHAPTER 32 / MASTER REFERENCE

APPENDIX B - early period content calendar

An editorial packet for the first fully interactive dates. The exact fictional examples are proposed content operations, while known releases remain clearly distinguished.

Summer 2012 source-led calendar

Date / window	Established anchor	Content build task
2012-07-11	Greg Episode 1: Before the Day Starts [EST]	Home PC, group chat, YAL summer threads, MeTube seed, park/store.
2012-07-12 morning	Episode 2 wake and group coordination [EST]	Message changes, PC history, optional music playback.
2012-07-12 midday	Book loan, mall, bookstore, food court [EST]	Book inventory, manga shelf, retail products, receipts.
2012-07-12 afternoon	Arcade and friend split [EST]	Racing, fighting, rhythm, score saves; Ren/Jay/Lexi routes.
2012-07-12 evening	Home PC, Volume 1, dinner and sleep [EST]	Reading state, Mom dialogue, night chat and saved profile.
2012-07 to 2012-12	Historical summer/fall internet [WORK]	Populate site archive incrementally without invented canonical events.
2012-09-01	Possible reboot announcement in earlier alternate draft [OPEN]	Block publication until date/sequence conflict is resolved.

2013 premiere-season chronological anchors

Date	Event	Archive behavior
2013-04-04	Iron to Requiem S1 Japanese premiere	ITR anime entry marked currently airing, episodic threads unlock.
2013-09-26	ITR S1 finale; IoE Episode 0 date	Two distinct works have distinct entries and sources.
2013-10-01	Chrome 30 stable [EXT]	Optional historic browser-chrome version available.
2013-10-02	Impact of Eden advance premiere	Advance event record, not regular TV broadcast.
2013-10-05	Impact of Eden regular TV premiere	Series status airing; event cascade to suitable sites.
2013-10-06 02:31 JST	YAL Episode 1 thread created	Thread visible from time; no later aggregates yet.
2013-10-09	Later YAL Episode 1 archive snapshot	1,113 replies and 42,806 views as snapshot, not creation count.
2013-10-13	YAL Episode 2 discussion dated	Episode 2 forum route becomes available.
2013-10-20	YAL Episode 3 discussion dated	Episode 3 forum route becomes available.

October 2013 day-by-day authored-content allocation

The following is a production checklist, not a claim that all listed fictional posts or videos historically existed. Each entry requires authored material and approval before entering the historical database.

2013-10-01 - editorial work item

Prepare Fall 2013 season homepage, ITR recap and current series grid. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-02 - editorial work item

Stage advance-premiere context and staff press dossier for IoE. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-03 - editorial work item

Complete artwork licenses and show proper IoE special vs premiere distinction. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-04 - editorial work item

Test Google-style search of prospective shows; prepare Saturday schedule. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-05 - editorial work item

Publish IoE premiere event, opening credits and relevant historical interface. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-06 - editorial work item

Create YAL ep1 thread at 02:31 JST, early comments and community links. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-07 - editorial work item

Add dated creator commentary and search results that reflect one aired episode. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-08 - editorial work item

Import a small authorized original AMV-inspired video and channel profile. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-09 - editorial work item

Capture YAL Episode 1 counts separately from live comments. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-10 - editorial work item

Refresh manga shop magazine shelf and audiovisual release bulletins. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-11 - editorial work item

Add period-correct forum signatures, reactions, typing style and user activity. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-12 - editorial work item

Produce a second original video review and corresponding discussion reference. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-13 - editorial work item

Open Episode 2 discussion at its established date; update links. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-14 - editorial work item

Review episode rankings and archive separate score snapshots. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-15 - editorial work item

Implement offline local shop, mall chatter and an event flyer tied to episode. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-16 - editorial work item

Test multi-tab browsing, side-by-side video and anime discussion. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-17 - editorial work item

Update search snippets and time filtering to include second week material. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-18 - editorial work item

Author a creator Q&A with a clearly constructed fictional publication record. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-19 - editorial work item

Release an original music clip with explicit rights metadata. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-20 - editorial work item

Open Episode 3 discussion at its established date. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-21 - editorial work item

Test forum page counts, quoted replies and long spoiler posts. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-22 - editorial work item

Publish authorized new artwork from fictional in-world fan accounts. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-23 - editorial work item

Add anime magazine issue in bookstore or gallery from authored assets. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-24 - editorial work item

Verify registered fictional websites keep independent identities. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-25 - editorial work item

Publish separate original comedy and gaming upload unrelated to IoE. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-26 - editorial work item

Run browser reconstruction fidelity and accessible keyboard checks. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-27 - editorial work item

Edit culture guide: original memes, slang and internet community behavior. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-28 - editorial work item

Create convention or local screening listing as clearly proposed reconstruction. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-29 - editorial work item

Capture snapshot metrics with explicit observation date and original source. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-30 - editorial work item

Test user who changes desktop wallpaper, logs off and returns. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

2013-10-31 - editorial work item

Reconcile October archive package; version and archive before November changes. Output must include at least one linked item ID, owner, approved source class, precise release-time field, and a future-leak regression test.

CHAPTER 33 / MASTER REFERENCE

APPENDIX C - reference screen anatomy and UI design contracts

Annotated screen layouts to support interface implementation without relying on a screenshot-only impression.

UI-01: 2012 boot and sign-in screen

Visual anatomy: Centered login/personal profile; wallpaper hint, login status; no futuristic assistant panel.

Essential real behavior: Session restore and privacy indicator. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-02: 2012 desktop

Visual anatomy: Background, compact shortcut grid, start menu, horizontal taskbar, clock; open app windows.

Essential real behavior: Drag, focus, minimize, task switch. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-03: Classic Fox browser frame

Visual anatomy: Older tab chrome, toolbar, address, back/forward, bookmark star, small menu.

Essential real behavior: Navigate internal/external and tabs. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-04: Classic Chromium browser frame

Visual anatomy: Tab shape, omnibox, star, compact settings, most visited new tab.

Essential real behavior: Search fallback, bookmark, restore. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-05: Search homepage

Visual anatomy: Large centered search, small utility links, optional image/news filters, no massive modern feed.

Essential real behavior: Instant server-backed result query. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-06: Search results

Visual anatomy: Ten-ish dense rows with title, origin, snippet, cached/archive affordance, pagination.

Essential real behavior: Date-aware ranked results and link open. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-07: MeTube homepage

Visual anatomy: Logo/top search; left navigation; featured and uploaded-video tiles; period-dense text.

Essential real behavior: Select video, channel, categories. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-08: MeTube watch page

Visual anatomy: Video on left, uploader meta and comments beneath, suggestions at right.

Essential real behavior: Functional player, likes, comments. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-09: YAL anime detail

Visual anatomy: Small cover/sidebar; title and condensed data row; tab controls; score, staff, synopsis.

Essential real behavior: List, rate, review and navigate. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-10: YAL episode thread

Visual anatomy: Dark top bar; crumb; teal title; metadata, horizontal action controls, 168px user pane and reply.

Essential real behavior: Spoiler, quote, pagination. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-11: Manga reader

Visual anatomy: Large central page, optional dual page, narrow chapter nav, discreet preferences.

Essential real behavior: Reader memory and page navigation. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-12: Messenger window

Visual anatomy: Friends at left or separate buddy list; message transcript; status dots; compact type field.

Essential real behavior: Presence, send, attachments. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-13: Birdie account

Visual anatomy: Small avatar/profile card, chronological posts, side rail of trends, short-character posts.

Essential real behavior: Follow, mention, repost, search. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-14: Tum blog

Visual anatomy: Large custom banner, tiled posts, archive by month, notes and reblogs.

Essential real behavior: Theme changes and media posts. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-15: 3Chan board

Visual anatomy: Light imageboard with reply links, thread counters, subject and image attachments.

Essential real behavior: Anonymous controlled posting. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-16: File Explorer

Visual anatomy: Folder path left tree; icons/list view; breadcrumbs; details bar; cut/copy/paste.

Essential real behavior: File CRUD. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-17: Downloads dialog

Visual anatomy: File name, source URL, size/progress, open or reveal target and retry buttons.

Essential real behavior: Persistent queued transfer. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-18: Local MP3 player

Visual anatomy: Library, now playing, playlist, seeking, volume, player skin optional.

Essential real behavior: Play background audio. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-19: Arcade menu

Visual anatomy: Diegetic machine screen, controls prompt, clear pause/retry, score display.

Essential real behavior: Play, record outcome. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-20: Mall directory

Visual anatomy: Map, store signage, opening hours, date-aware stock and available activities.

Essential real behavior: Navigate venues and browse. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-21: Content admin

Visual anatomy: Source list, conflicts, publish actions, effective date and revision history.

Essential real behavior: Approve/import and audit. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

UI-22: Historical clock picker

Visual anatomy: Calendar and optional time zone, visible historic vs modern action label.

Essential real behavior: Switch session snapshot safely. Component states: empty, loading, success, unauthorized, offline and not-yet-published. Keyboard focus visible; small historical text remains zoomable.

Historical proof: capture a reference image for the selected year, verify relevant release date, annotate asset provenance and attach an approval stamp before release.

CHAPTER 34 / MASTER REFERENCE

APPENDIX D - product names, domains and legal separation

Terminology for real browsers, search engines, fictional equivalents, website branding and archived interfaces.

Identity and treatment matrix

Historical reference	Connected Era deliverable	Mode and restriction
Google Search	Connected Search engine with period-indexed content	Original service; modern Google optionally launches outside historical mode.
Mozilla Firefox	Classic Fox-style browser interface	Original branded shell look; never call it an official Firefox binary.
Google Chrome	Classic Chromium-style browser interface	Modern secure renderer with 2013 visual chrome.
Internet Explorer	Optional historical third interface theme	Only if it adds period value; modern security preserved.
YouTube	MeTube	Original fictional video platform; no unauthorized video mirroring.
MyAnimeList	YAL / YourAnimeList	Distinct fictional database styling, names, content and account system.
Crunchyroll-like services	MunchyRoll	Fictional streaming platform; original or licensed media only.
Tumblr-like blogging	Tum	Independent blog service with its own features.
Twitter-like microblog	Birdie	Period-specific 140-character social service.
4chan-like imageboards	3Chan	Anonymous boards with contemporary moderation controls.
Flash websites	Original HTML5 games or Ruffle-managed SWFs	Emulation only for permitted content, not all Flash.
Wayback Machine	Read-only external historical archive integration	Never treat missing links/scripting as fully working [W10].
Windows 7	2012 Connected Desktop shell	Aesthetic reference, not redistributed Microsoft OS.

Original artwork & media asset strategy

- Commission or author new assets for animated series, video thumbnails, music visualizers, desktop icons, wallpapers and fictional merchandise.
- Every asset has rights_owner, proof reference, license type, publication date, supported platforms and media rating.
- Third-party service logos are navigational references only where necessary and permitted; avoid passing off fictional services as real company products.

- Use publicly available reference screenshots to study geometry and conventions, not as entire copied application pages bundled inside the product.
- Prioritize the exact YAL HTML already in the project as a design source, while maintaining independent final branding.

How the initial 2012-13 browsing mix feels

Greg can open the original Connected Search homepage, a Fox-style browser, YAL, MeTube, a manga reader, a general forum, a messaging client, RSS reader, music player, local downloads, gaming sites and personal blogs. A real modern Google site remains available as an explicitly modern external link when the user leaves the historical simulation mode.

CHAPTER 35 / MASTER REFERENCE

APPENDIX E - definitive acceptance and provenance checklist

Final static audit for the design book before the engineering team begins implementation.

- CHECK-01: Every named application family has a real functional contract.
- CHECK-02: Every user-visible action has an expected persistent or observable result.
- CHECK-03: Every app supports working empty/loading/error states.
- CHECK-04: Every original website resolves from stable internal URLs.
- CHECK-05: External live web restrictions are documented and securely handled.
- CHECK-06: Every planned content class has an owner and licensing field.
- CHECK-07: All historical fictional records carry source ID and revision.
- CHECK-08: Unresolved canon conflicts are registered and have safe interim behavior.
- CHECK-09: Time gate depends on published_at and appropriate jurisdiction/time zone.
- CHECK-10: Historic captured metrics are not misrepresented as earlier counts.
- CHECK-11: Modern user content cannot masquerade as old archived records.
- CHECK-12: Data model has stable IDs and correct one-to-many relationships.
- CHECK-13: YAL episodes, characters, composers, teams and studio credits are linked.
- CHECK-14: YAL visual palette and 2013 thread anatomy are documented.
- CHECK-15: IoE October 2013 day-specific reveal tests are specified.
- CHECK-16: ITR 102-episode update is reflected in the chronology.
- CHECK-17: EARTH: Metal Comet cross-studio seasons are separate records.
- CHECK-18: Flow and Burst filmographies remain individually indexed.
- CHECK-19: LUNE and other music roles have distinct vocal and composer credit types.
- CHECK-20: OFFLINE day 1 to day 2 persistence has functional tests.
- CHECK-21: World town name/alternate cast are not prematurely fixed.
- CHECK-22: Manga reading state respects the physical borrowed book story.
- CHECK-23: MeTube video upload and player quality/rights are specified.
- CHECK-24: Browser runtime sandbox and permission handling are specified.
- CHECK-25: File storage and upload security have threat mitigations.
- CHECK-26: Privacy deletion/export and moderation policy have required endpoints.
- CHECK-27: Accessibility acceptance is referenced to WCAG 2.2.
- CHECK-28: All external technical historical facts have source links.
- CHECK-29: The first build remains scoped to 2012-13 despite 2006-26 master architecture.
- CHECK-30: The PDF explicitly distinguishes a complete specification from completed software.

Specification handoff milestone

On delivery of this master PDF: 100% of the identified document domains and implementation contracts are described. This is a documentation milestone only. Engineering, content production, copyright clearance, source reconciliation and deployment have independent completion gates.